HYDRIS AI

HYDRIS FIELD GLASSES · PRODUCT REQUIREMENTS

Product Requirements Document

What Hydris Field Glasses must do for wastewater operators, why, how we will know it works, and what ships when

DOCUMENT	VERSION	STATUS	DATE
HFG-PRD	2.0	Draft for review	October 2026

Document control

Document	Product Requirements Document (HFG-PRD)
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Status	Draft for review
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Source of truth	docs/product/PRD.md in tarun-km/hydrisview (this Word file is generated from it)
Classification	Confidential: Hydris, Inc. and named partners

Revision history

Version	Date	Author	Changes
2.0	October 2026	Product, Hydris	Expanded edition. Re-baselined every requirement against main (07cca23), added the native Android app, acceptance criteria, MoSCoW priorities, journeys, lifecycle diagrams, KPIs, release plan and traceability

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This document is the single statement of what Hydris Field Glasses is for and what it must do. It is written for the people who build the product, the people who sell and deploy it, and the plant teams who will use it. Every statement about the current product was checked against the repository on `main` at commit `07cca23` (22 October 2026). Where the code and an earlier document disagree, the code wins and this document says so.

Each requirement carries a stable ID, a MoSCoW priority, a status and an acceptance criterion. Status labels mean exactly this:

- **Built**: implemented on `main` and exercised by the product today (in the live demo or in automated tests).
- **Partial**: implemented in part, or implemented in code but not yet validated where it matters (for example on physical Ray-Ban Meta glasses).
- **Planned**: not in the repository. It is a commitment to build, with a target phase in the release plan.

Targets (latency, accuracy, adoption) are labelled as targets. None of them is a measured result unless the text names the measurement.

1 Executive summary

Wastewater plant operators spend their shift walking the plant: reading instruments, checking pumps and blowers, sampling effluent, writing values on paper logs, and responding to alarms that usually reach a control room before they reach the person standing next to the equipment. The records they produce are the evidence a plant shows to its regulator, and much of that evidence still lives on paper.

Hydris Field Glasses puts the live state of the plant, the next task and the approved procedure in the operator's line of sight, hands free, and turns every reading, check and paper record they touch into a trustworthy digital record. It does this through five connected surfaces that share one API, one tenant model and one permission matrix:

1. A **public website** with a live, synthetic glasses demo.
2. An **operator console** for supervisors, managers and administrators.
3. A **glasses heads-up app** (web) that runs on smart-glasses runtimes, in a phone layout, and inside a browser simulator.
4. A **native Android app**, Hydris Field, which connects Ray-Ban Meta glasses through Meta's Wearables Device Access Toolkit (DAT 1.0.1).
5. A **multi-tenant API** with row-level security, realtime events and an append-only audit trail.

67
automated tests passing
26 domain + 41 API, run locally on 22 Oct 2026

5
product surfaces
website, console, glasses app, Android app, API

143
API routes
across 15 modules in apps/api/src/modules

12
glasses screens
UI_SCREENS in packages/domain/src/uiprofile.ts

6
roles
38 permissions in packages/domain/src/roles.ts

16
measurands
pH to residual chlorine, with unit normalization

42
database tables
23 enums, RLS on every tenant table

10
demo scenarios
deterministic, demo tenants only

The platform core is built and running at `https://hydrisview.vercel.app` in demo mode (named guest accounts on synthetic plant data). The main gaps before a customer pilot are: validation of the glasses paths on physical Ray-Ban Meta hardware, email delivery for invitations and password resets, continuous integration, a scheduled-job cadence that suits real data on serverless hosting, and an offline outbox in the Android app. The release plan (chapter "Release plan and milestones") sequences these into a hardening phase, a pilot at one or two Indian plants in Q1 2027, and general availability in Q2 2027.

DECISION

Hydris is read-only towards the plant. It never writes to PLCs, SCADA or equipment. The grounded assistance endpoint says so in every response ("Hydris does not control equipment", `apps/api/src/modules/insights.ts`), and the voice assistant's tools are all read-only (`apps/api/src/modules/voice-assistant.ts`). This keeps Hydris outside the plant's control-system safety case and is a precondition for every requirement in this document.

IMPORTANT

Ray-Ban Meta glasses reach third-party apps only through Meta's DAT, and the glasses are paired, registered and permissioned through the Meta AI app. Hydris cannot replace the Meta AI app; it must stay installed on every operator phone. The Android app checks for it (`GlassesManager.metaAiInstalled()`, package `com.facebook.stella`) and sends the user there to connect.

2 Vision and product principles

2.1 Vision

Every operator in a water or wastewater plant sees what matters, does the next right thing, and leaves behind a record that stands up to an audit, without putting down their tools.

In three years we want Hydris to be the default way an Indian ETP, STP or CETP runs its rounds, responds to alarms and keeps its compliance records, and a credible choice for plants in other markets with similar discharge-monitoring obligations.

2.2 Product principles

These principles settle design arguments. When two requirements conflict, the principle higher in this list wins.

Table 1. Product principles and how the code already applies them

#	Principle	What it means in practice	Where the code applies it
P1	Safety first, read-only towards the plant	Never issue a control command. Point to the approved procedure and the supervisor for safety-critical actions	`/assist` safety notice; assistant system prompt in `voice-assistant.ts`
P2	Never show a guess as a value	Missing data is "no data", never zero. Stale values are marked and excluded from health roll-ups	`freshness()` in `packages/domain/src/freshness.ts`
P3	Every number has provenance	A reading says where it came from (instrument, manual, lab, simulator) and when	`TELEMETRY_SOURCES`, DB check `telemetry_synthetic_source_ck`
P4	History is never overwritten	Corrections append; the original stays visible	Telemetry corrections, audit document `extracted` values, `audit_events`
P5	Humans verify, machines assist	AI and OCR propose; a person confirms; a different person approves	Audit document four-eyes rule in `audit-docs.ts`
P6	Glance, then act	A screen on the glasses answers one question. Voice answers are one or two sentences	HUD screens; assistant `SYSTEM` prompt
P7	Hands busy, still in control	Every operator action works with directional input or voice; the camera only runs on explicit intent	`useHudInput` in `apps/web/src/glasses/input.ts`; scan screens
P8	Tenant isolation is enforced by the database, not by convention	RLS on every tenant table; every request runs as `hydris_app` with `app.org_id`	`drizzle/0001_rls.sql`, `db/tenant.ts`
P9	Synthetic data can never reach a customer tenant	Simulator readings are refused for non-demo organizations	`ingest()` in `modules/telemetry.ts`
P10	One account everywhere	The same user, permissions and data on console, glasses and phone	Shared API; Android `HydrisApi.kt` uses the web session contract

2.3 What Hydris is not

- Not a SCADA, DCS or historian. It reads from them (Planned connectors) and never replaces them.
- Not a general-purpose chatbot. The assistant answers only from the tenant's own data and approved documents.
- Not a laboratory information management system. It records lab results as telemetry with `source = lab` and will integrate with a LIMS later (Planned).
- Not a consumer glasses app. It is an enterprise tool distributed to managed devices.

3 Problem and market context

3.1 The operator's day today

Effluent treatment plants (ETPs), sewage treatment plants (STPs), water treatment plants (WTPs) and common effluent treatment plants (CETPs) run continuously, with shift operators walking fixed rounds. The facility kinds Hydris models are exactly these (`FACILITY_KINDS` in `packages/domain/src/catalog.ts`). A typical round covers intake and screening, equalization, aeration, clarification, filtration, disinfection, sludge handling and the discharge point, which are the process areas Hydris models (`PROCESS_AREA_KINDS`).

Four problems recur:

1. **Paper rounds drift from reality.** Readings are written on clipboards, transcribed later, and diverge from instrument values. Nobody can tell which value was observed and which was copied.
2. **Alarms reach the wrong person.** Alarms land in a control room or on a panel. The operator nearest the pump finds out last. Acknowledging an alarm and actually fixing the condition get conflated, so the record shows alarms "cleared" that were never resolved.
3. **Compliance evidence is unsearchable.** Logbooks, lab sheets, calibration certificates and inspection checklists are paper. Producing evidence for an inspection means digging through files.
4. **Phones are the wrong tool at the asset.** Gloves, ladders, wet hands and sampling bottles make phones awkward. Smart glasses free the hands, but there is no industrial software for them.

3.2 Regulatory context

Hydris is designed around discharge-monitoring regimes, and it states only what is well established about them:

- **India.** Discharges are regulated under the Water (Prevention and Control of Pollution) Act, 1974, through consents to establish and operate issued by State Pollution Control Boards (SPCBs) and Pollution Control Committees. General discharge standards are set under the Environment (Protection) Rules, 1986, with plant-specific limits in each consent. In 2014 the Central Pollution Control Board (CPCB) directed the 17 categories of highly polluting industries, and CETPs, to install online continuous effluent and emission monitoring systems (OCEMS) connected to the CPCB and SPCB servers (CPCB online monitoring).
- **United States.** Point-source discharges need National Pollutant Discharge Elimination System (NPDES) permits, with monitoring results reported through Discharge Monitoring Reports.
- **European Union.** The Urban Waste Water Treatment Directive (originally 91/271/EEC, recast in 2024) sets treatment and monitoring requirements for urban wastewater.

The common thread is that plants must monitor continuously, keep records that can be retrieved and trusted, and show who did what and when. Hydris does not certify compliance. It produces records (telemetry with provenance, inspection results, verified audit documents and an append-only audit trail) that make compliance easier to demonstrate.

> **NOTE**
>
> Hydris does not replace OCEMS reporting to CPCB or SPCB servers, and it is not a certified OCEMS. A plant's OCEMS analyzers can feed Hydris through the gateway ingestion API (FR-022) so operators see the same values the regulator sees.

3.3 Why glasses, why now

Three things changed. Consumer-grade camera glasses (Ray-Ban Meta) are light, affordable and socially acceptable on a plant floor. Meta opened a device access toolkit (DAT) that lets third-party Android and iOS apps use the glasses' camera, microphone, capture button and speakers. And large language models can now answer spoken questions from structured plant data reliably enough to be useful, provided they are grounded and constrained (chapter "Data and AI requirements").

3.4 Competitive landscape (qualitative)

Table 2. Alternatives a plant uses today and where Hydris differs

Alternative	Strength	Gap Hydris fills
Paper logbooks and clipboards	Familiar, no devices needed	No provenance, no search, no realtime alarms, transcription errors
SCADA/HMI in the control room	Authoritative live data and control	Not at the asset; no rounds, no paper-record digitization
Generic mobile forms apps	Cheap digital checklists	Not hands-free, no live telemetry context, no alarm lifecycle, no verified OCR
Remote-assist video apps for glasses	Expert sees what the operator sees	No plant data model, no rounds, no records
CMMS / EAM suites	Strong maintenance planning	Heavy, desktop-first, weak in the field, no glasses

4 Personas

Hydris has seven personas. Six map one to one onto the roles in `packages/domain/src/roles.ts`; the platform administrator is a flag on the user (`users.is_platform_admin`).

Table 3. Personas, roles and primary surfaces

Persona	Role in code	Primary surface	Top needs
Field operator (Ravi)	`field_operator`	Glasses, Android app	Next task, live values at the asset, acknowledge alarms, record readings, scan paper logs, ask by voice
Shift supervisor (Meena)	`supervisor`	Console, Android app	Dispatch rounds, triage and assign alarms, review inspections, verify scanned documents
Maintenance engineer (Arjun)	`maintenance_engineer`	Android app, glasses	Work orders, checklists, investigate and resolve alarms, SOPs at the asset
Plant manager (Priya)	`factory_manager`	Console	KPIs and reports, alarm rules, approve audit documents, devices and people
Compliance auditor (Mr Rao)	`viewer` (read plus `report.read`)	Console, PDF exports	Tamper-evident trail, approved documents, exports for the regulator
Organization admin (IT lead)	`org_admin`	Console	Members and invitations, sites, glasses UI profiles, all permissions
Hydris platform admin	`users.is_platform_admin`	Console `/console/platform`	Create and suspend customer organizations

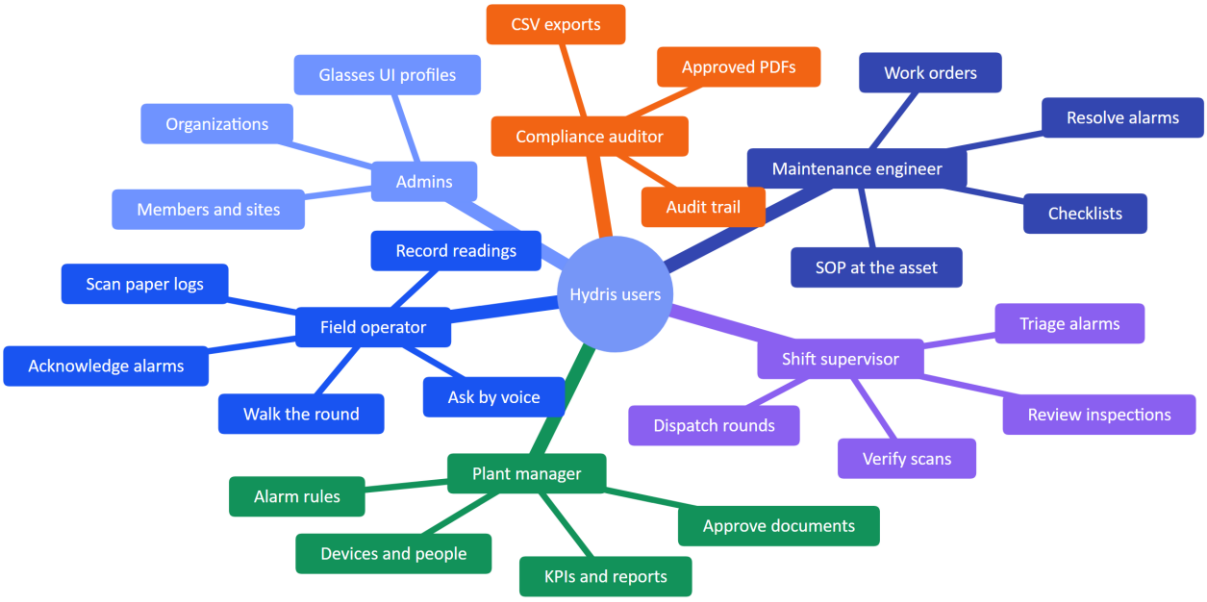


Figure 1. *Personas and their jobs to be done*

4.1 Persona detail

Field operator

Ravi is 28, a diploma holder with six years on ETP shifts. He works 8-hour rotating shifts, often alone at night. He wears gloves and safety glasses, carries a sampling bottle and a radio, and has an Android phone in his pocket. His English reading is good; he speaks Hindi and Tamil at home. He wants fewer forms, clear instructions, and to be told quickly when something near him is wrong. He fears being blamed for a reading he copied from someone else.

What success looks like for Ravi: his round is on the glasses in order, each asset shows its live values with how fresh they are, he records a reading with two taps or a sentence, and his paper log becomes a verified PDF without retyping.

Shift supervisor

Meena runs a shift of four operators across two plants. She lives in the console during the day and on her phone at night. She needs to see open alarms by severity, who has them, and which rounds are late. She verifies scanned audit documents against the paper before her manager approves them.

Maintenance engineer

Arjun moves between pumps, blowers and dosing systems. He needs the work order, the checklist and the equipment manual at the asset, and he needs to close the loop on alarms: investigate, fix, resolve with a note.

Plant manager

Priya is accountable to the SPCB for discharge compliance and to her management for uptime and cost. She configures alarm rules, approves audit documents, decides which glasses features each role sees, and reads the reports.

Compliance auditor

Mr Rao is an internal or third-party auditor. He needs read-only access to approved records, the verification trail of every scanned document, and exports. He must never be able to change anything.

Administrators

The organization admin manages members, invitations, sites and glasses UI profiles. The Hydris platform admin creates and suspends customer organizations and has no access to tenant data by default.

5 Jobs to be done and user journeys

5.1 Jobs to be done

Table 4. Jobs to be done, with the requirements that serve them

#	Job statement	Served by
J1	When I start my round, I want my assigned inspections in order, so I do not miss an asset	FR-040, FR-041, FR-059
J2	When I stand at an asset, I want its live readings and how fresh they are, so I can trust what I record	FR-021, FR-026, FR-095
J3	When an alarm fires near me, I want to acknowledge it and see the recommended action and SOP, so the right person is accountable	FR-031, FR-032, FR-033, FR-112
J4	When I fill a paper log or receive a lab sheet, I want to capture it once and have it verified, so the record is searchable and defensible	FR-060 to FR-069
J5	When my hands are busy, I want to ask by voice and hear a short answer grounded in our data	FR-071, FR-072, FR-074
J6	As a manager, I want to control what each role sees on the glasses without shipping an app update	FR-054
J7	As an admin, I want to pair, reassign, suspend or revoke a pair of glasses in seconds	FR-050 to FR-053, FR-057
J8	As an auditor, I want every approved document and every change traceable to a person and a time	FR-065, FR-066, FR-082
J9	As a prospect, I want to try the product in minutes without talking to sales	FR-083, FR-091, FR-102

5.2 Journey: an operator's shift

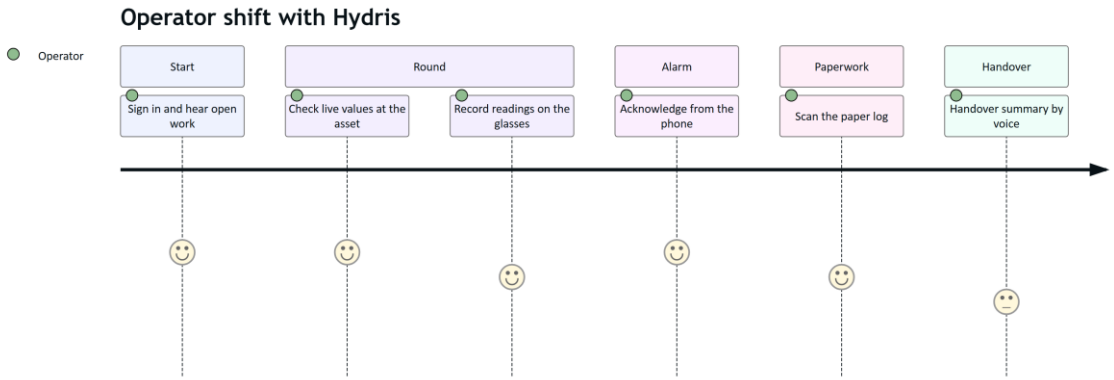


Figure 2. *Field operator journey across one shift (satisfaction 1 to 5, target state)*

The shift journey shows where the product is strong today (alarm acknowledgement, scanning, voice lookups) and where it is weakest: the handover summary is a Planned capability, and inspection execution on the Android app is Planned (FR-044). Inspections are executed on the glasses heads-up app today.

5.3 Journey: from paper record to approved evidence

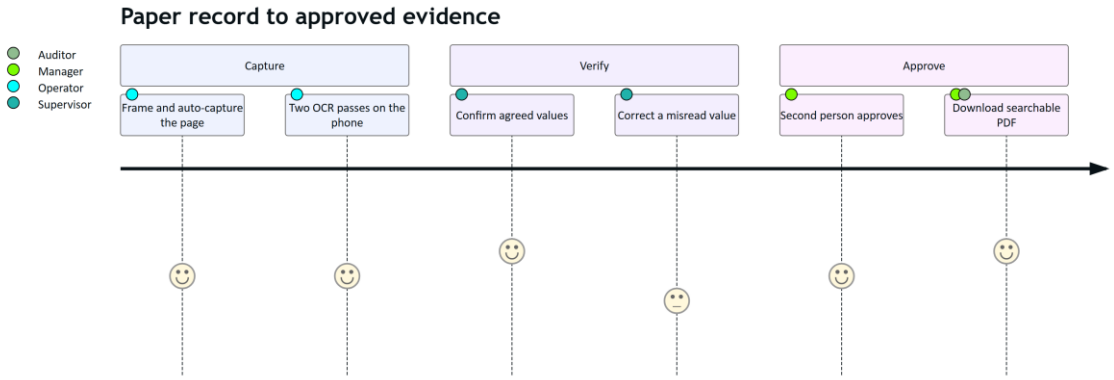


Figure 3. *Supervisor and manager journey for a scanned audit document*

5.4 Journey: evaluating Hydris

A prospect lands on the website, opens the live glasses demo (`/glasses?demo=1`, fully client-side synthetic data), and then signs in to the console or the Android app in demo mode with just their name (`POST /auth/demo`). Each guest gets an `org_admin` account in the demo tenant with personal copies of the demo owner's scheduled inspections and open maintenance tasks (`apps/api/src/core/demo-guests.ts`). Nothing they do affects other guests' work, and all data is clearly labelled synthetic.

6 Product surfaces and scope

6.1 System context

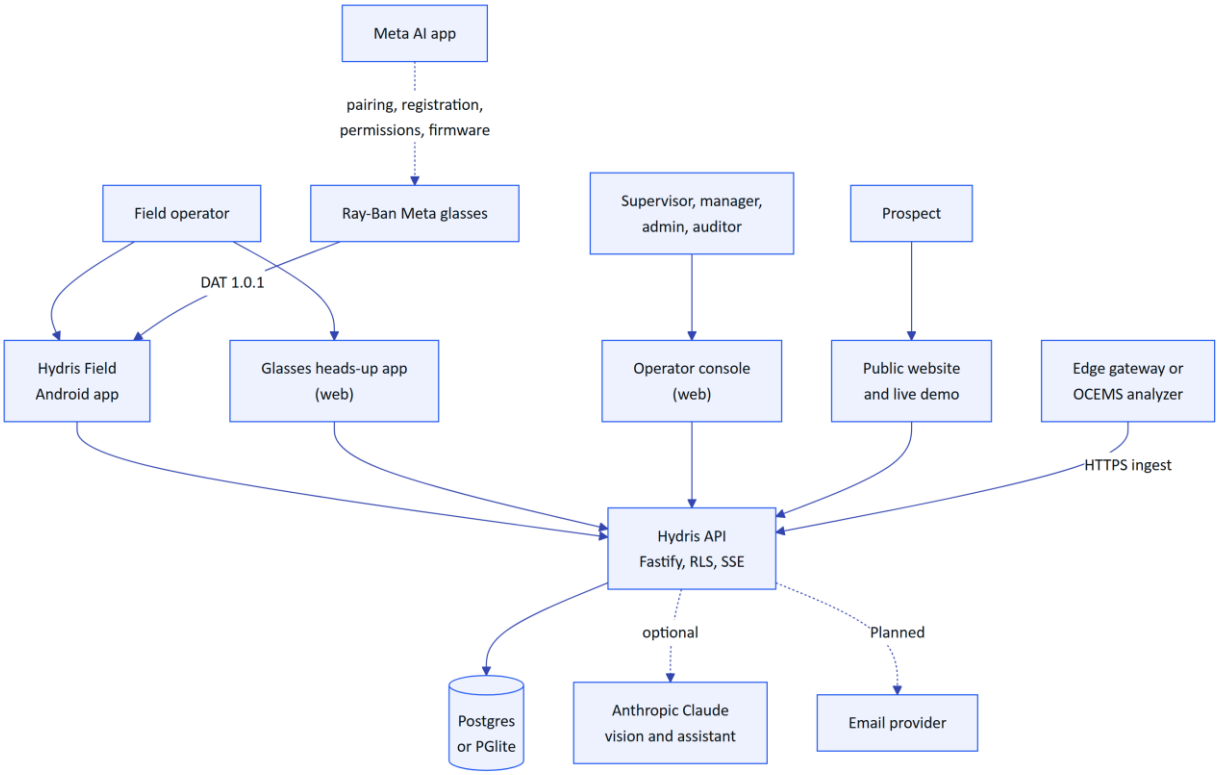


Figure 4. *System context: people, Hydris surfaces and external systems*

6.2 Surfaces

Table 5. Product surfaces, where they live and their status

Surface	Code	What it does	Status
Public website	`apps/web/src/app/(site)`	Seven pages (`/`, `/product`, `/technology`, `/operations`, `/demo`, `/contact`, `/privacy`), live glasses demo, contact and demo-request form	Built
Operator console	`apps/web/src/app/console/**`	Sixteen navigation sections in five groups (Operate, Work, Plant, People and devices, Insight) plus Admin, account and platform pages	Built
Glasses heads-up app	`apps/web/src/app/glasses`, `apps/web/src/glasses/*`	Twelve screens, directional input, offline inspection queue, audit scan, voice assistant, phone layout	Built (web). Native glasses display rendering Planned
Simulator and pairing	`/simulator`, `/pair`	Browser glasses simulator; approve a pairing code by QR or typed code	Built

Surface	Code	What it does	Status
Android app "Hydris Field"	`apps/android`	Sign-in, home, work, alarms, sensors, audit scan with ML Kit OCR, "Hey Hydris", glasses connection and registration through DAT	Partial: built in code, glasses paths not validated on hardware
Android WebView bridge	`apps/glasses-bridge-android`	Hosts the glasses web app and bridges DAT camera, speech, capture button and voice invocation	Partial: superseded by the native app for pilot
API	`apps/api`	143 routes in 15 modules, RLS tenancy, SSE realtime, jobs, demo mode	Built

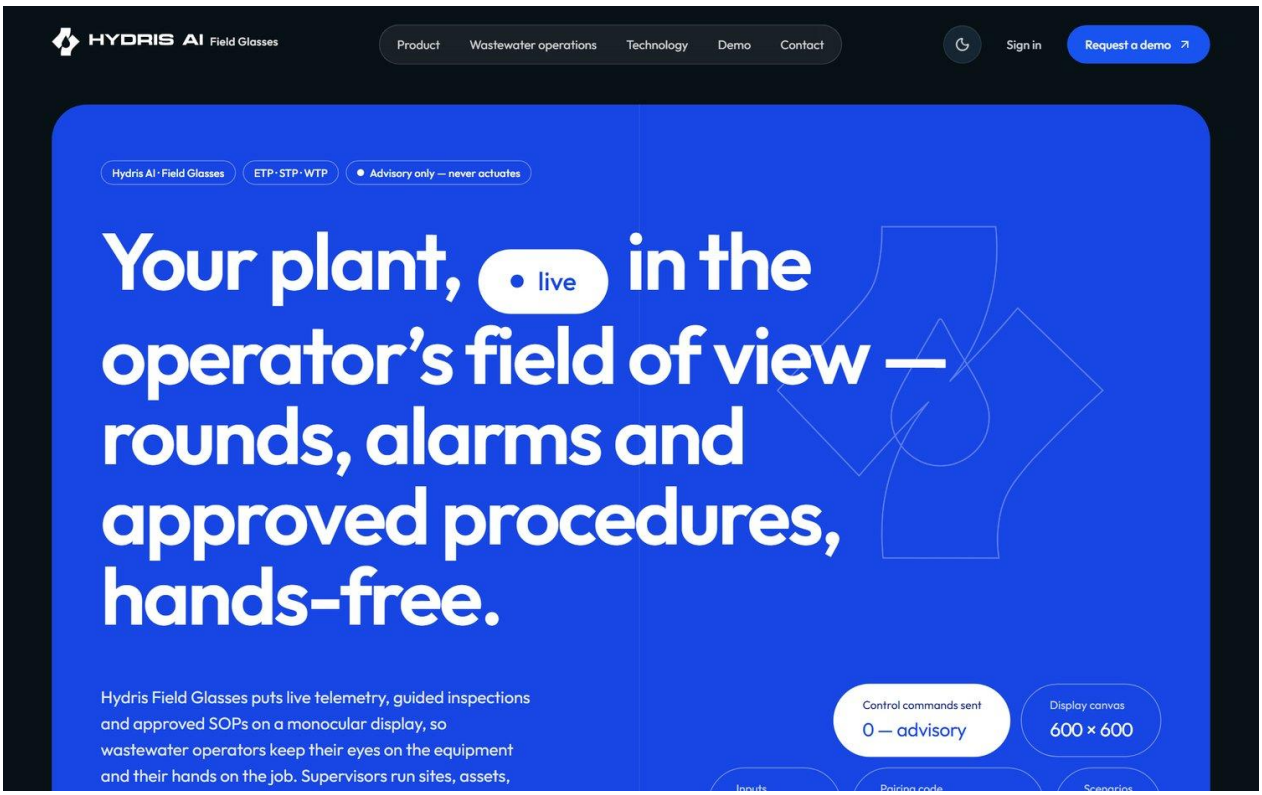


Figure 5. *The public website home page*

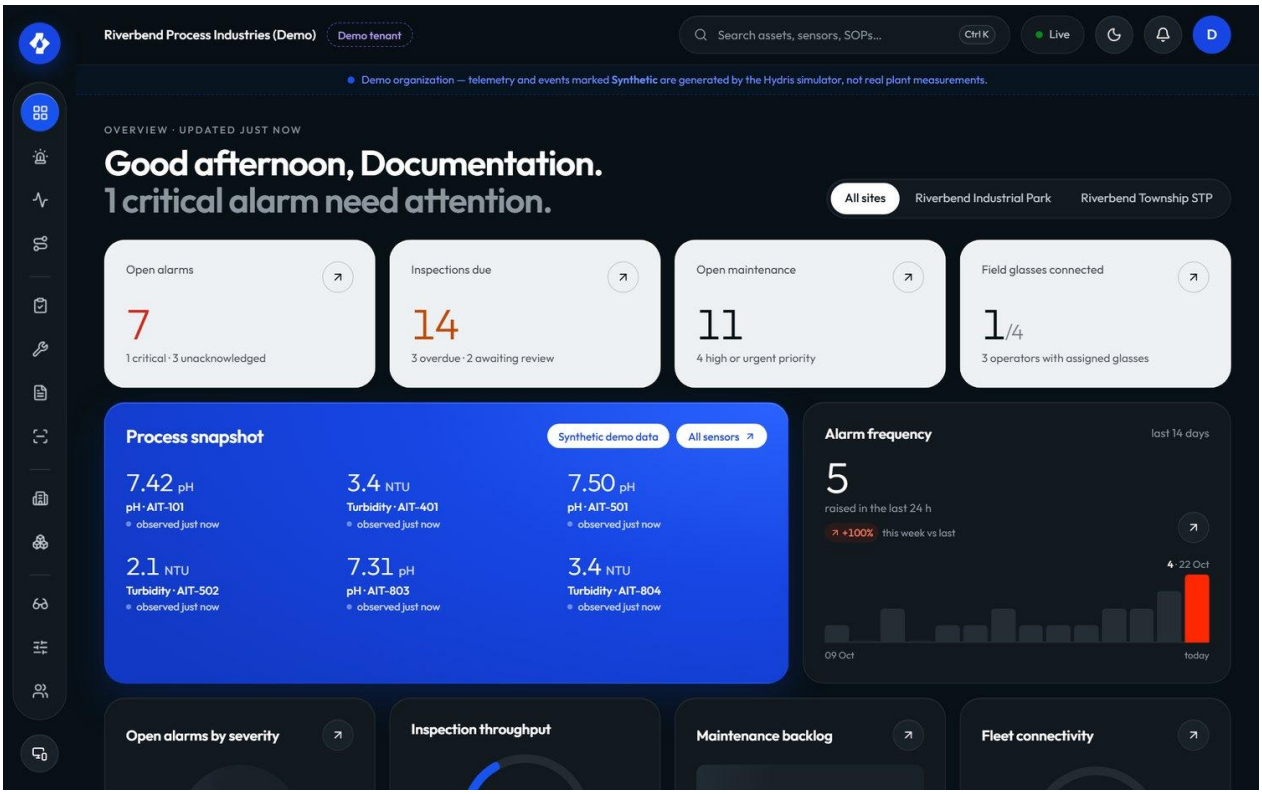


Figure 6. *The operator console overview: KPIs, open alarms and live measurements*

6.3 Feature map

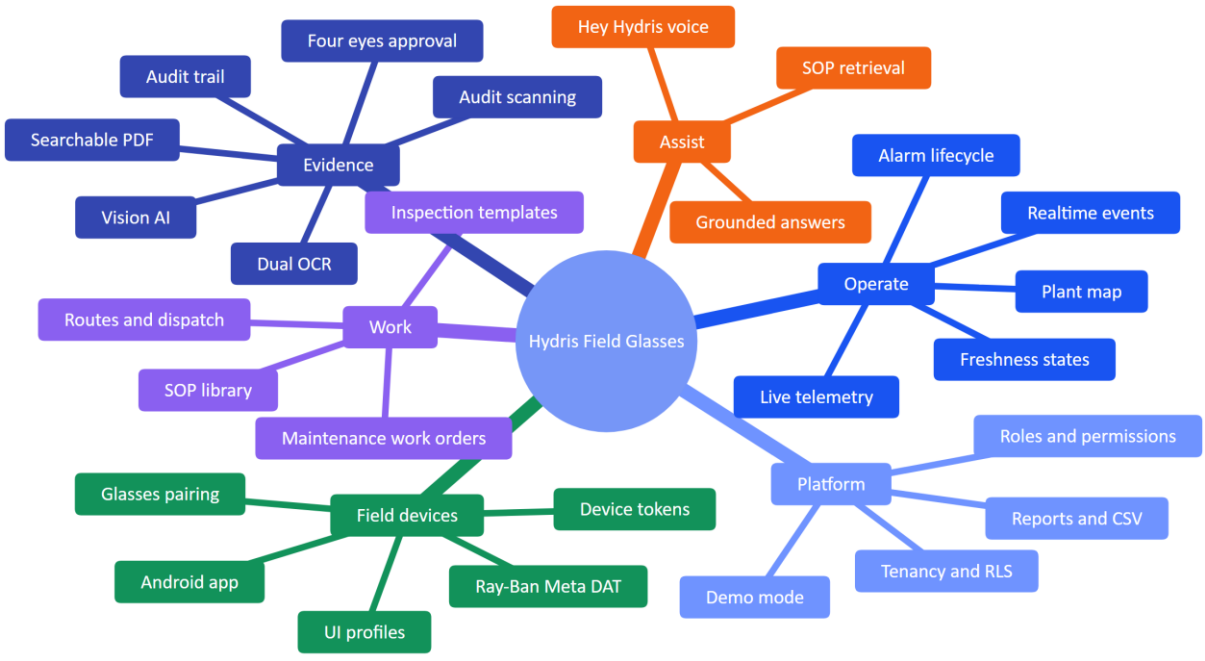


Figure 7. *Feature map of the platform by capability area*

6.4 In scope and out of scope

Table 6. Scope boundaries for the pilot and GA releases

Area	In scope	Out of scope
Plant control	Read-only display of values, alarms and procedures	Any write to PLC, SCADA or equipment; setpoint changes
Data ingestion	HTTPS gateway push, manual and lab entry; MQTT and OPC UA adapters (Planned)	Direct fieldbus integration (Modbus, Profibus) inside Hydris
Devices	Ray-Ban Meta through DAT on Android; glasses web app on other runtimes and in a phone layout	Building custom glasses hardware; iOS before GA
Compliance	Records with provenance, verification and audit trail; exports	Certified OCEMS reporting to CPCB or SPCB servers; legal e-signature certification before FR-068
AI	Grounded answers from tenant data; vision extraction with human verification	Autonomous decisions; free-form generation of procedures; answers from the open internet
Accounts	Invitations, MFA, sessions, SSO (Planned)	Self-serve signup for customer tenants (demo mode only)
Languages	English UI and voice	Hindi, Tamil, Marathi and Gujarati voice (Planned, Advanced phase)

7 Functional requirements

7.1 How to read the requirement tables

Each table lists ID, requirement, MoSCoW priority (**M**ust, **S**hould, **C**ould, **W**on't this release), status, acceptance criterion and the surfaces involved (W website, C console, G glasses app, A Android app, P API). IDs from version 1.0 are unchanged. New IDs take the next free number inside the area's block of ten; new areas start a new block (FR-090 Android app, FR-100 website and demo, FR-110 documents and SOPs, FR-120 insights and compliance).

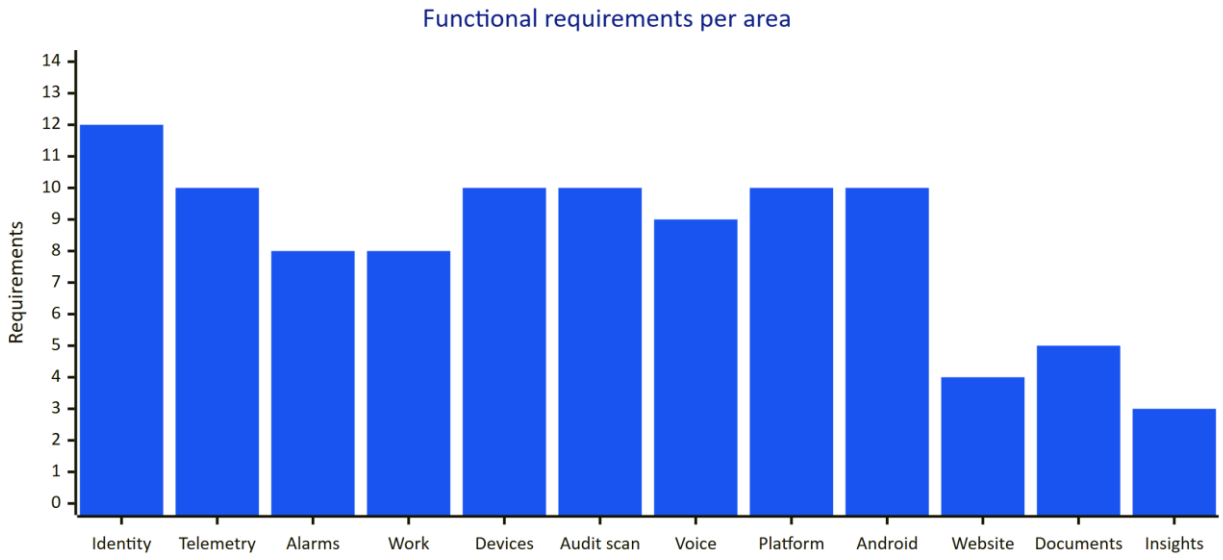


Figure 8. *Requirements per functional area in this edition*

7.2 Identity, tenancy and access

Hydris is multi-tenant from the first line of code. Every tenant table has row-level security; every request runs `SET LOCAL ROLE hydris_app` with the organization in `app.org_id`, so the database enforces isolation even if application code has a bug. Roles are a fixed matrix of six roles and 38 permissions; the API enforces it on every request and clients use it only to decide what to show.

Table 7. Identity, tenancy and access requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-001	Email and password sign-in with lockout after 5 failed attempts for 15 minutes	M	Built	Sixth attempt within the window returns `423 account_locked`; lockout clears after 15 min (`TTL.lockoutMs`)	C, A, P
FR-002	TOTP MFA (setup, enable, disable, verify step) with the secret encrypted by `DATA_KEY`	S	Built	MFA user cannot obtain a session without a valid code; secret is never stored in clear	C, P
FR-003	Refresh-token rotation with family revocation on reuse (30 s multi-tab grace)	M	Built	Reusing a rotated refresh token outside the grace window revokes the whole family	C, G, A, P

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-004	Invitations with a 7-day token and role assignment that cannot escalate above the inviter (`assignableRoles`)	M	Built (delivery depends on FR-086)	A factory manager cannot invite an org admin; expired tokens are refused	C, P
FR-005	Password reset with a 30-minute single-use token	M	Partial	Token works once and expires in 30 min. Gap: the email is written to `mail_outbox` and not sent	C, P
FR-006	Multi-organization membership and organization switch	S	Built	A user in two orgs switches with `/auth/switch-org` and sees only the new org's data	C, P
FR-007	Site-scoped permissions for every role except `org_admin`	M	Built	A site-scoped supervisor cannot read alarms of another site (403/404)	All
FR-008	Database-enforced tenant isolation (RLS)	M	Built	Cross-tenant reads return no rows even with a crafted ID (API test suite)	P
FR-009	Session listing and revocation	S	Built	Revoking a session ends it on the next refresh	C, P
FR-010	Platform organization management (create, suspend)	M	Built	Suspended org members cannot sign in to that org	C, P
FR-011	Single sign-on with SAML 2.0 and OIDC per organization	S	Planned (GA)	An org admin configures an IdP; users sign in with it; local passwords can be disabled	C, A, P
FR-012	Fixed role matrix: six roles, 38 permissions, no escalation path	M	Built	`can(role, permission)` is the only check; unit tests cover the matrix	All

> **NOTE**
> The Android app does not yet complete an MFA challenge. If an account has MFA enabled, the app tells the user to sign in on the web console first (`LoginScreen.kt`). That gap is tracked as FR-099.

7.3 Plant model and telemetry

The plant model is a hierarchy: site, facility (ETP, STP, WTP, CETP, other), process area (13 kinds), asset (14 types), and sensors on assets. Sensors measure one of 16 measurands with defined units and decimals (`MEASURAND_DEFS`). Readings carry a source (`gateway`, `manual`, `lab`, `simulator`) and a quality. Freshness is derived from the sensor's expected interval: live within 2 times the interval, delayed within 5 times, stale beyond, and `no_data` when nothing was ever received.

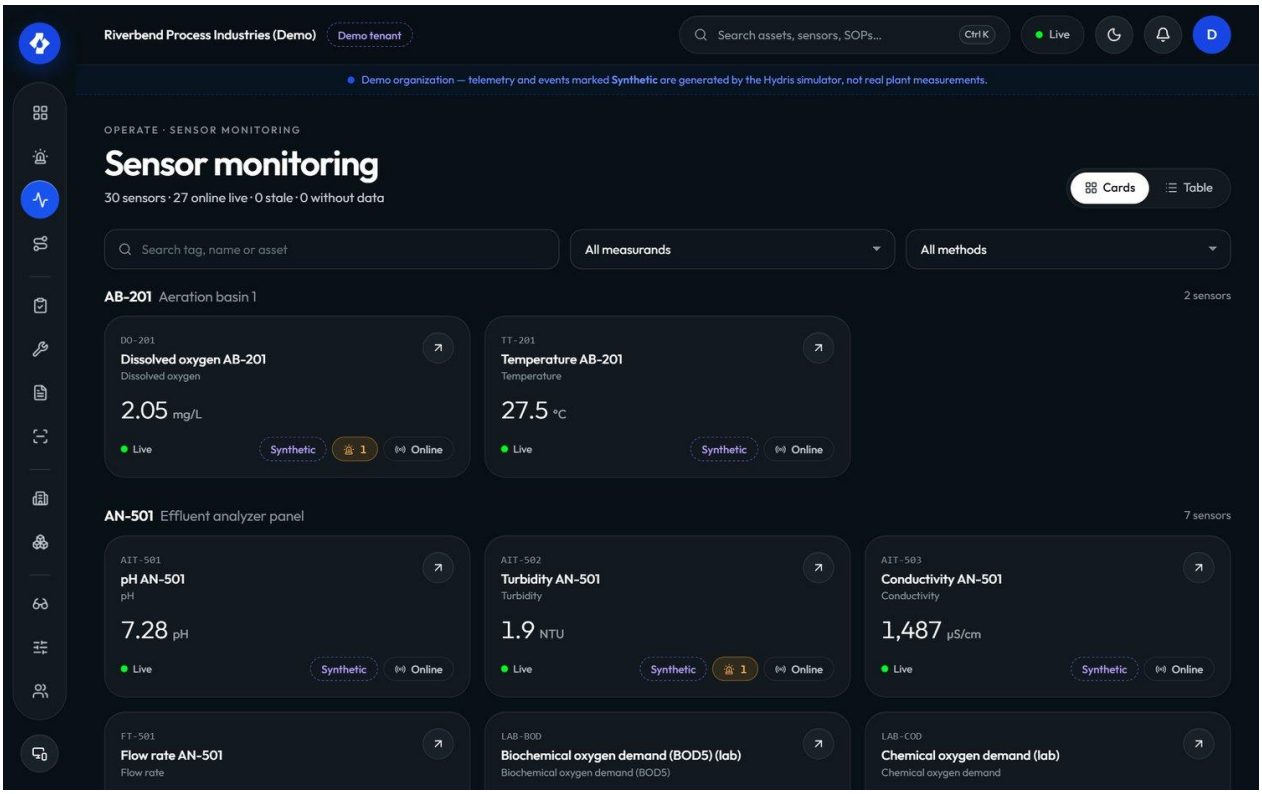


Figure 9. *Sensor monitoring in the console: live values with freshness and provenance*

Table 8. Plant model and telemetry requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-020	Site, facility, process area and asset hierarchy with map positions, archive and restore	M	Built	Archived assets disappear from glasses and assistant answers and can be restored	C, G, P
FR-021	Sensors for 16 measurands with unit normalization	M	Built	A reading in a compatible unit is stored in the canonical unit	C, P
FR-022	HTTPS gateway ingestion `POST /ingest/v1/readings` with an `hgw_` API key, up to 500 readings per call, idempotent on sensor and timestamp	M	Built	Replaying the same batch creates no duplicates; a revoked key gets 401	P
FR-023	Manual and lab entry with provenance	M	Built	Entries show source "Manual" or "Lab" everywhere they appear	C, P
FR-024	Append-only telemetry corrections (correct or invalidate)	M	Built	The original reading stays visible with its correction and author	C, P
FR-025	History, raw readings and CSV export per sensor	S	Built	CSV export opens in a spreadsheet without formula injection (leading `=+-@` escaped)	C, A, P
FR-026	Freshness states live, delayed, stale, no data; missing data never shown as a value	M	Built	A sensor with no readings shows "No data", never 0	All
FR-027	MQTT and OPC UA ingestion adapters	S	Planned (Advanced)	A gateway publishing over MQTT or OPC UA lands readings with the same guarantees as FR-022	P
FR-028	Read-only SCADA and historian connectors	C	Planned (Advanced)	Values from a historian appear with source and timestamp; no write path exists	P

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-029	Synthetic data is labelled everywhere and refused for non-demo tenants	M	Built	Simulator readings into a customer org are rejected with `synthetic_not_allowed`; DB check ties `synthetic` to `source = simulator`	All

7.4 Alarms

Alarms come from threshold rules (high-high, high, low, low-low) with a deadband (hysteresis) and an on-delay, or from a stale-data rule. An alarm has a severity (critical, high, medium, low) and moves through a five-state lifecycle. Acknowledging records that a person has seen it; it does not change the process. Resolving records that the condition was dealt with and requires a note. Closing archives the event after review.

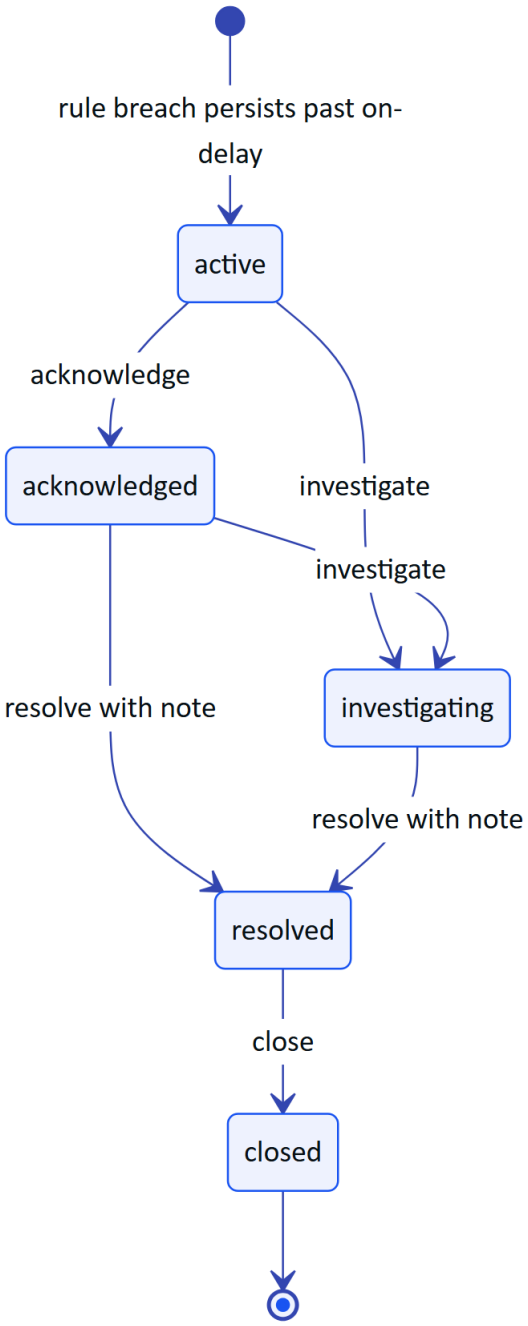


Figure 10. *Alarm lifecycle (packages/domain/src/alarms.ts)*

Table 9. **Alarm transitions, permissions and who holds them**

Action	From	To	Permission	Note required	Held by
Acknowledge	active	acknowledged	`alarm.acknowledge`	No	All roles except viewer
Investigate	active, acknowledged	investigating	`alarm.investigate`	No	Admin, manager, supervisor, maintenance engineer
Resolve	acknowledged, investigating	resolved	`alarm.resolve`	Yes	Admin, manager, supervisor, maintenance engineer
Close	resolved	closed	`alarm.close`	No	Admin, manager, supervisor

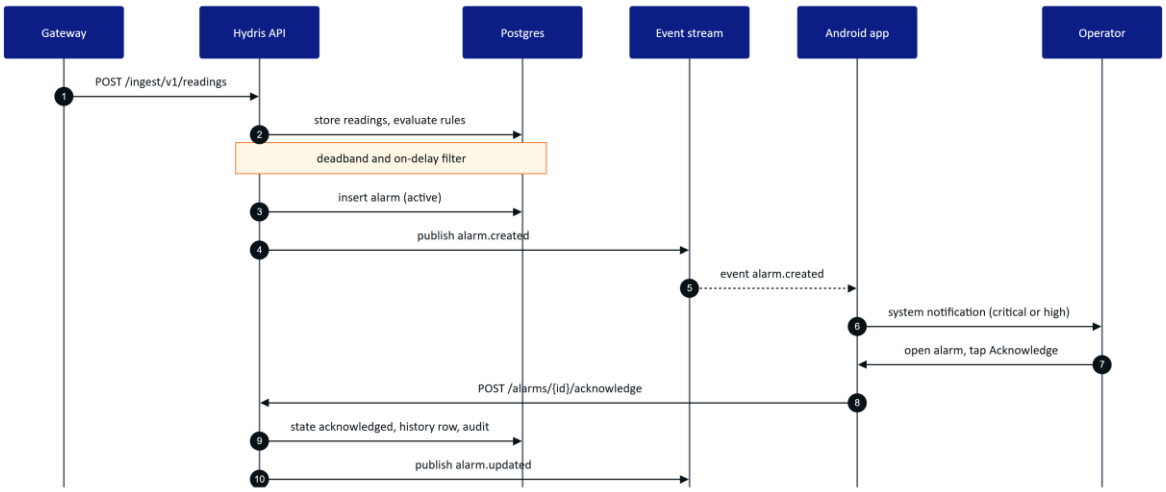


Figure 11. *From a sensor reading to an acknowledged alarm on the operator's phone*

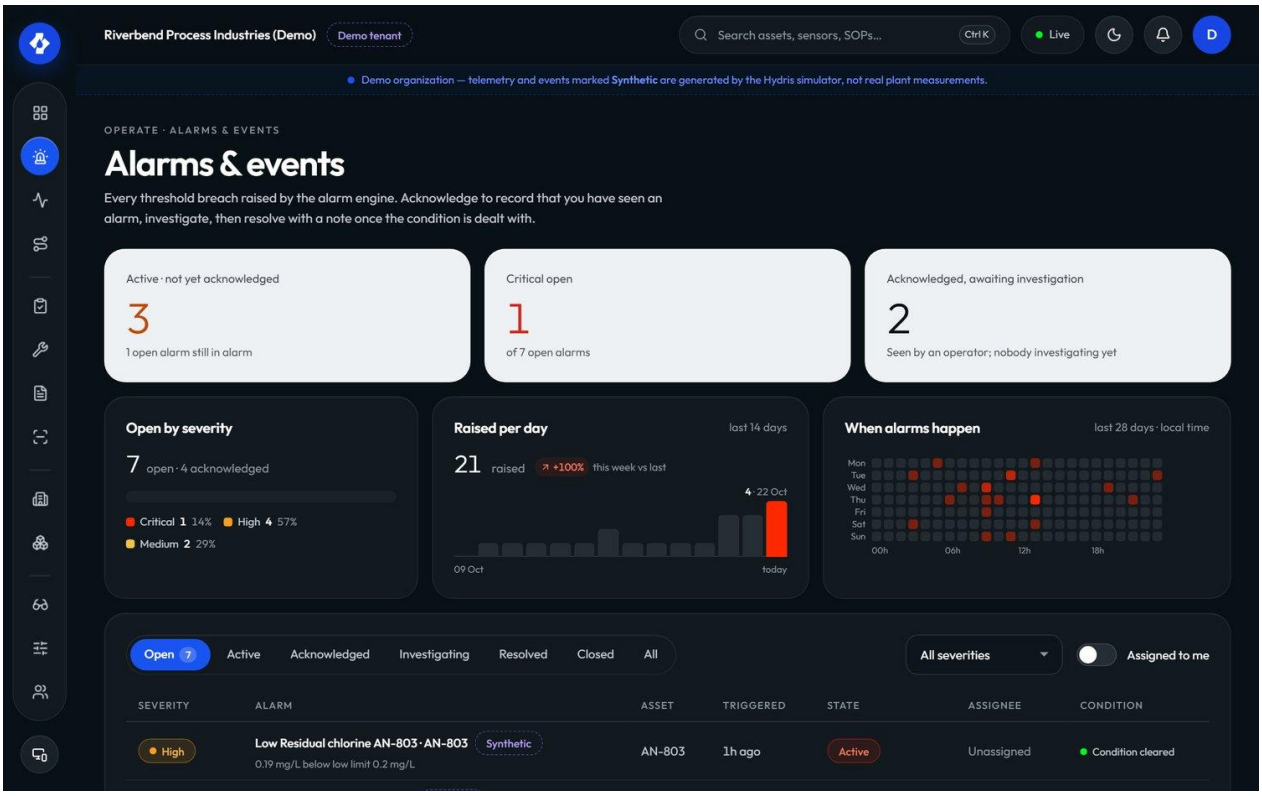


Figure 12. *Alarms and events in the console*

Table 10. Alarm requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-030	Threshold rules (high-high, high, low, low-low, stale) with deadband and on-delay	M	Built	A value oscillating inside the deadband does not raise or clear repeatedly; a breach shorter than the on-delay raises nothing	C, P
FR-031	Lifecycle active, acknowledged, investigating, resolved, closed; no resolve from active; resolution note required	M	Built	`POST /alarms/{id}/resolve` without a note returns `400 note_required`	C, G, A, P

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-032	Assignment, notes, history, recommended action and linked SOP	M	Built	Alarm detail shows history with actors, the rule's procedure and follow-up tasks	C, A, P
FR-033	Realtime push of alarm events to console, glasses and Android	M	Built	A new alarm appears without reload within the NFR-002 target	C, G, A, P
FR-034	Push notifications through Firebase Cloud Messaging when the app is not running	S	Planned (GA)	A critical alarm reaches a phone whose app process is stopped	A, P
FR-035	Stale-data alarms evaluated on a schedule	M	Partial	Long-running server evaluates every 30 s (`startJobs`). Gap: on Vercel the cron runs once a day (`vercel.json`, `0 3 * * *`)	P
FR-037	Escalation when an alarm stays unacknowledged (notify supervisor after a configurable time)	S	Planned (Pilot)	An unacknowledged critical alarm notifies the site supervisor after N minutes; the escalation is in the alarm history	C, A, P
FR-038	Alarm shelving with reason and expiry	C	Planned (GA)	A shelved alarm is hidden from the operator's active list until expiry; shelving is audited	C, P

NOTE

FR-036 from the draft of this edition (Android alarm notifications) was folded into FR-097 to keep Android requirements in one block. The ID is retired and will not be reused.

7.5 Inspections and maintenance

Inspection templates are versioned and made of steps of five kinds: check, reading (with expected range), choice (with abnormal options), text and photo. Routes group templates in order and are dispatched to operators. An inspection moves through scheduled, in progress, paused, submitted, then approved or returned (rejected), or cancelled. A reading outside its expected range, a failed check or an abnormal choice marks the step abnormal automatically (`stepOutcome`).

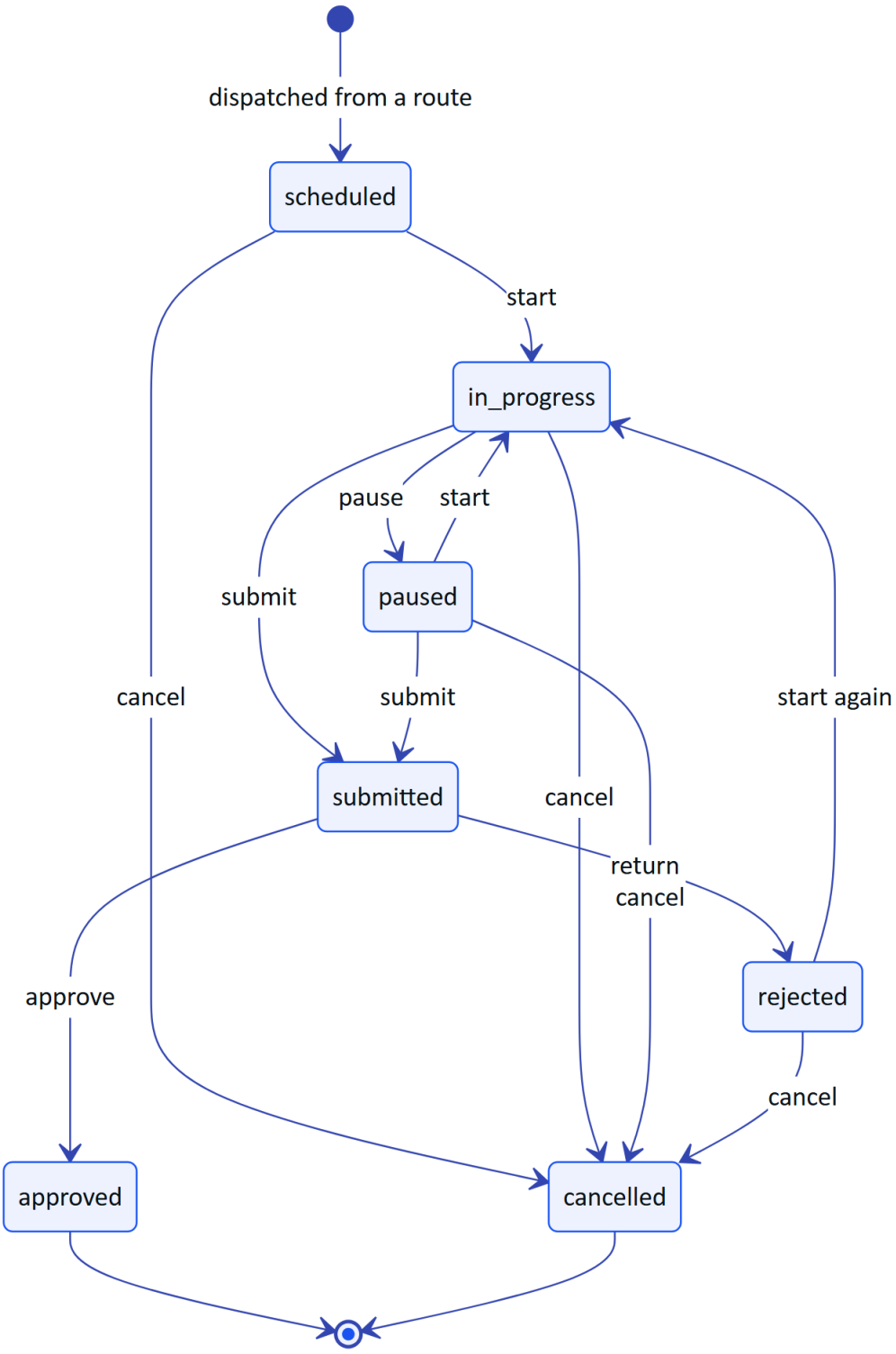


Figure 13. *Inspection lifecycle (packages/domain/src/workflows.ts)*

On the glasses, every step response is queued locally first and then flushed; replays are idempotent because each response carries a client operation ID (`apps/web/src/glasses/data.ts`). An operator who loses connectivity mid-round keeps working and the record syncs when the network returns. The "Glasses lose connectivity" demo scenario (`glasses_network_loss`) exercises exactly this.

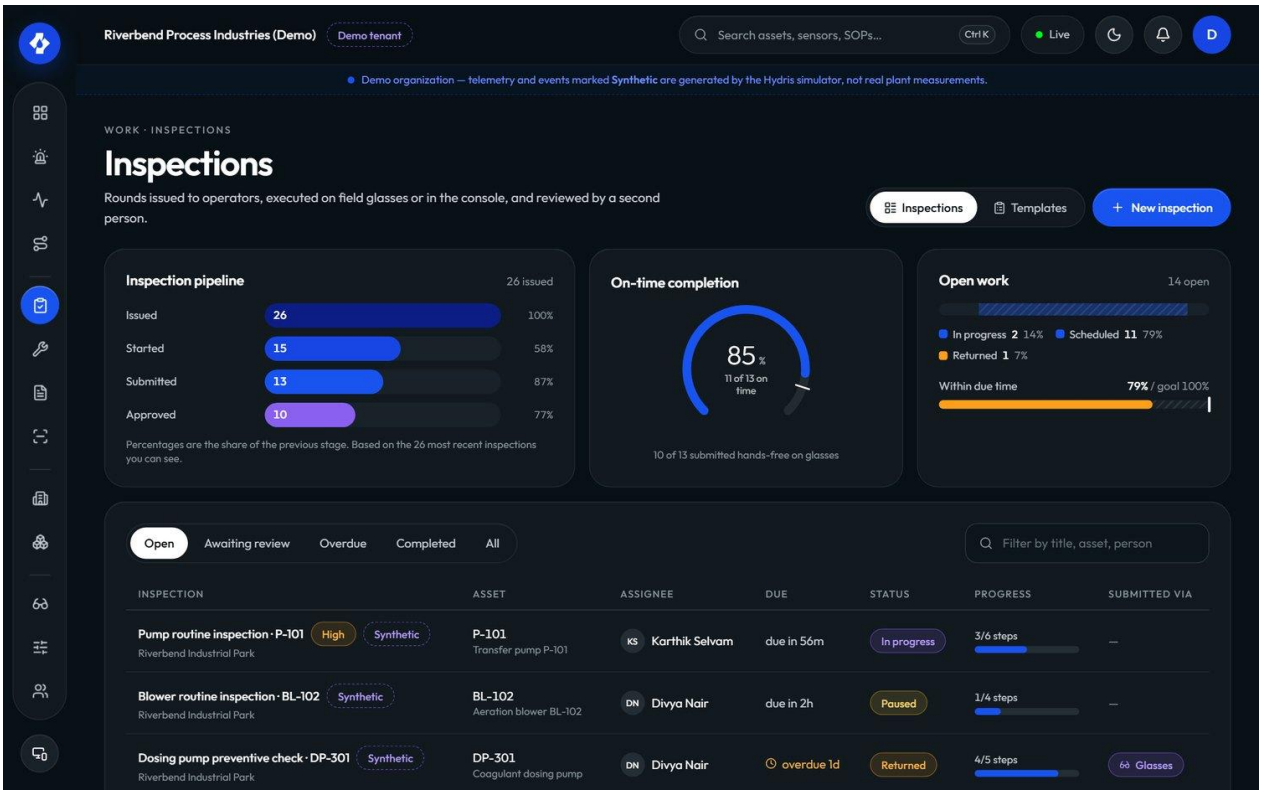


Figure 14. *Inspections in the console*

Maintenance tasks are corrective, preventive, inspection follow-ups or alarm follow-ups, with a priority (low to urgent), a checklist and an activity log. Their lifecycle is open, assigned, in progress, on hold, completed or cancelled, with reopen from completed or cancelled.

Table 11. Inspection and maintenance requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-040	Versioned inspection templates (check, reading, choice, text, photo) and routes with dispatch	M	Built	Editing a template creates a new version; dispatched inspections keep the version they were created with	C, P
FR-041	Execution on the glasses with an offline queue and idempotent replay	M	Built	Responses recorded offline appear once after reconnect (Playwright journey)	G, P
FR-042	Submit, review, approve or return	M	Built	A returned inspection can be restarted and resubmitted; history shows both submissions	C, G, P
FR-043	Maintenance tasks with checklist, activity and follow-ups from alarms and inspections	M	Built	A follow-up created from an alarm links back to it and shows on the alarm detail	C, A, P
FR-044	Inspection execution in the Android app (step entry, submit)	S	Planned (Pilot)	An operator completes a round on the phone with the same validation and offline guarantees as FR-041	A, P
FR-045	Photo steps captured with the glasses or phone camera	S	Partial	Photo steps and attachments exist (`/attachments`); the glasses app shows "attach from console" because camera capture is not wired to photo steps	G, A, P
FR-046	Tick maintenance checklist items from the Android app	M	Built	Ticking an item on the phone updates the console task in realtime	A, P

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-047	Recurring preventive maintenance and inspection schedules (calendar or runtime based)	S	Planned (GA)	A weekly route generates inspections automatically; a runtime-hours rule creates a task when crossed	C, P

7.6 Devices and glasses

A pair of glasses becomes a Hydris device through a pairing flow shaped like the OAuth 2.0 Device Authorization Grant (RFC 8628). The glasses ask for a challenge and display a short code and a QR link; a signed-in person approves it on another device and chooses the site and operator; the glasses, holding a separate poll secret that is never displayed, then claim a device credential. The QR code never contains a credential.

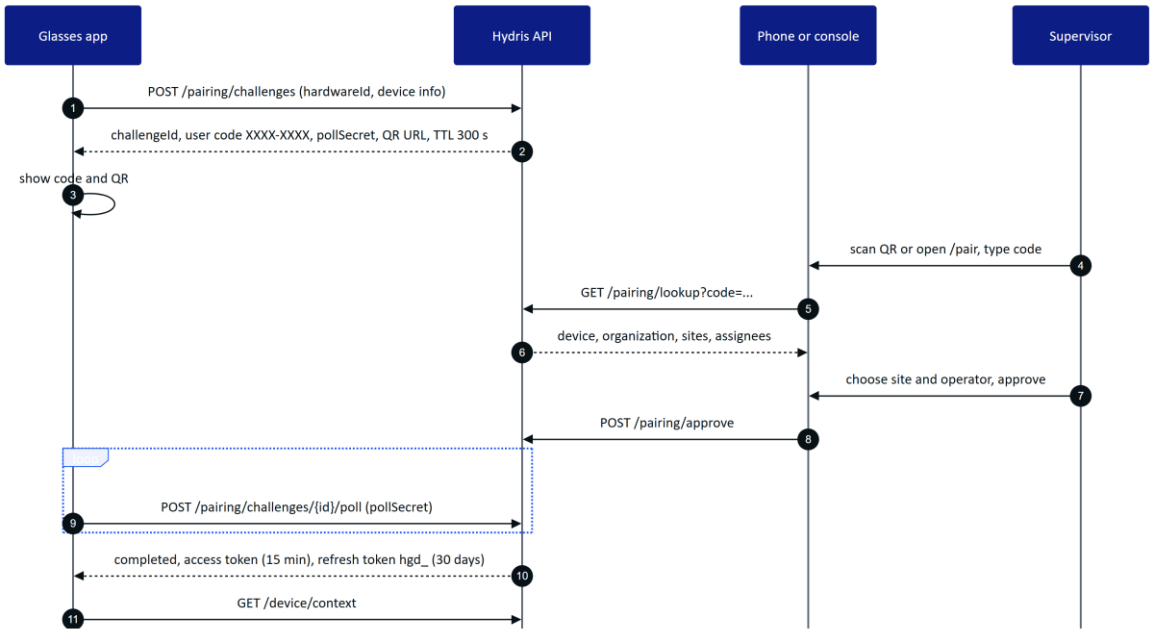


Figure 15. *Glasses pairing, an RFC 8628-style device flow (apps/api/src/modules/devices.ts)*

Device credentials are short-lived access tokens (15 minutes) plus a rotating 30-day refresh token with reuse detection. A session version on the device record lets an administrator invalidate every outstanding token at once. Suspending, revoking or unassigning a device takes effect on its next request and on the realtime stream's 30-second revalidation.

Device permissions are capped: whatever the assigned operator's role, a device can only do what `DEVICE_PERMISSIONS` allows. Managers further shape the glasses experience with **UI profiles**: versioned configurations of which screens are enabled and in what order, which home widgets show, which measurements are allowed and in which order per asset type, which actions are permitted, and layout (density, text scale, handedness). The server enforces the same configuration it sends to the device.

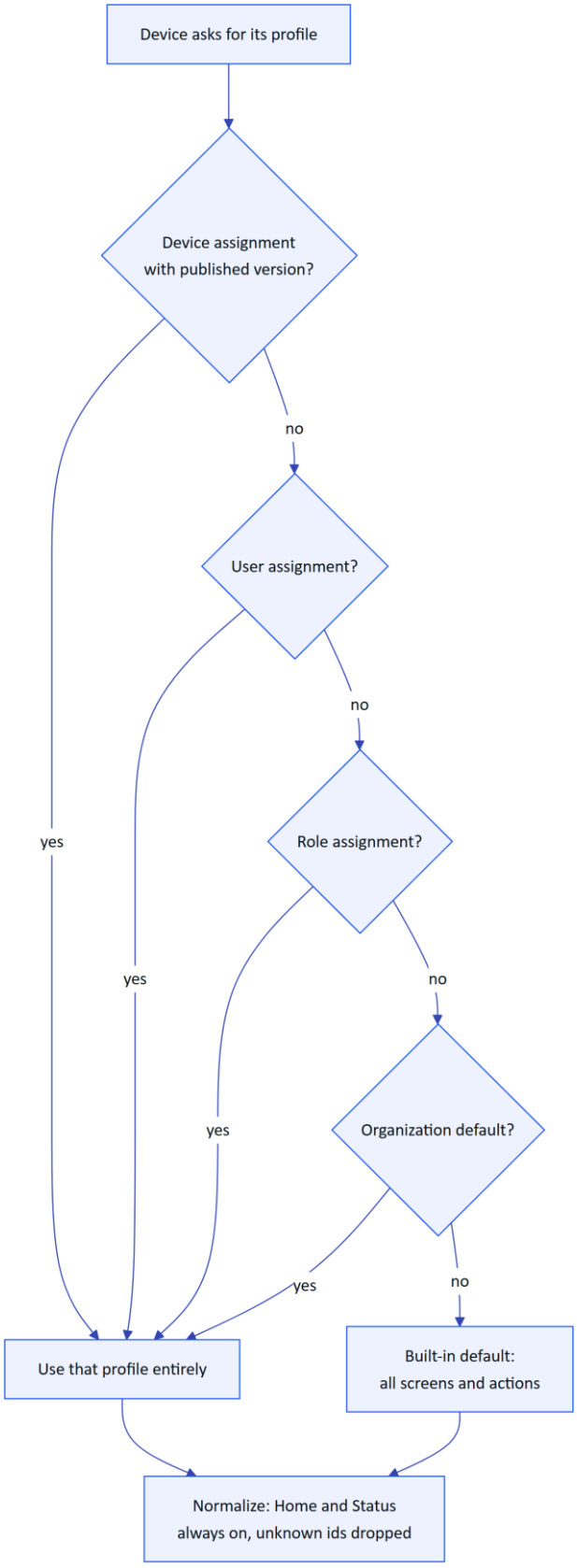


Figure 16. *How the effective glasses UI profile is resolved (packages/domain/src/uiprofile.ts)*

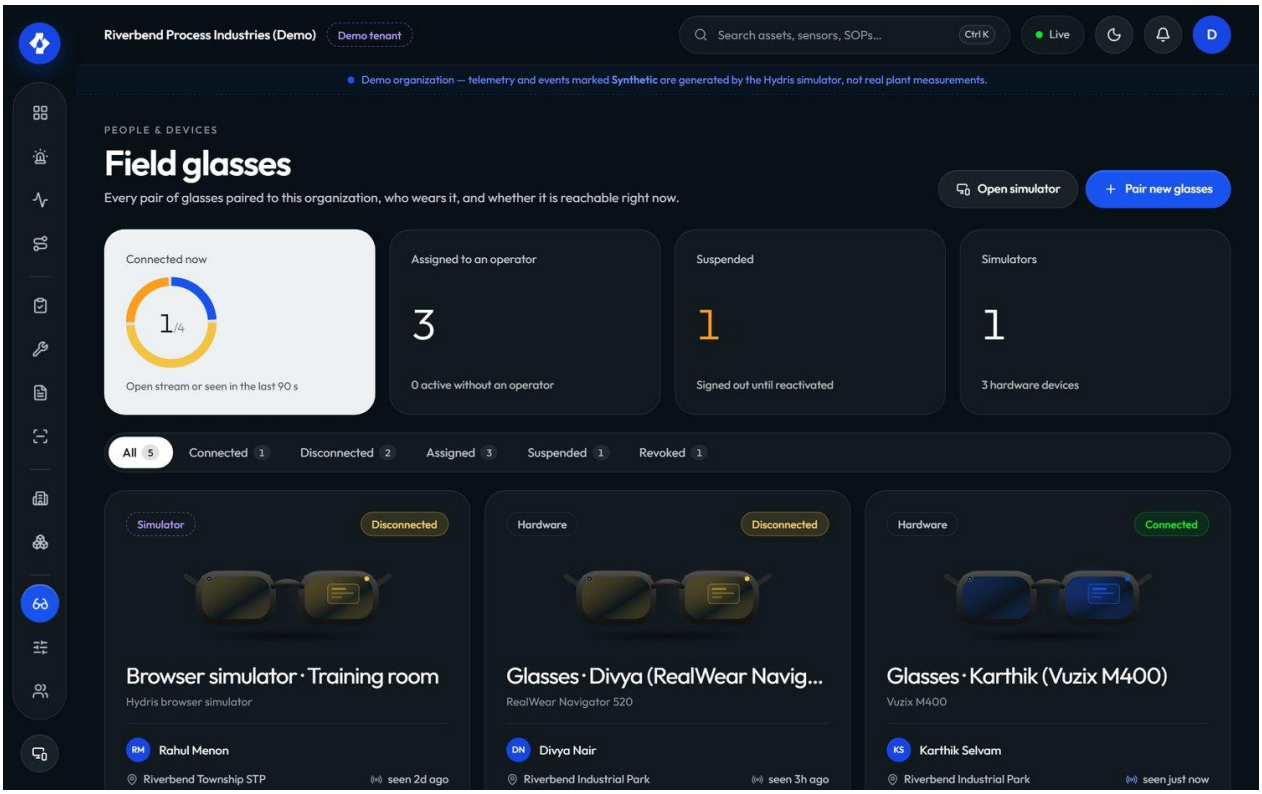


Figure 17. *Field glasses in the console: pairing state, assignment and last seen*

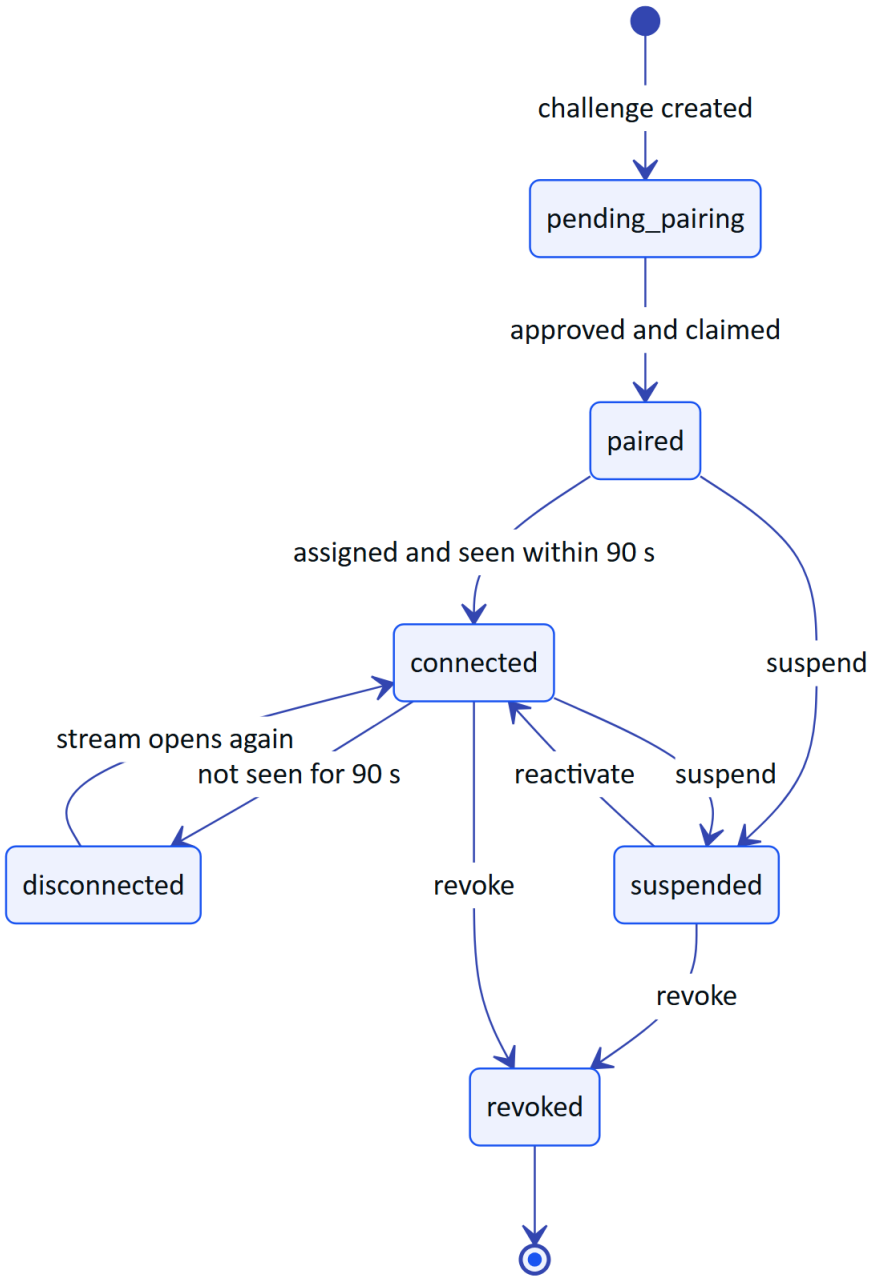


Figure 18. *Composite device states shown to administrators (packages/domain/src/devices.ts)*

Table 12. **Device and glasses requirements**

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-050	Pairing with an 8-character code (no 0, O, 1, I, L, U), QR link, 5-minute TTL, 3 s polling, approval with site and assignee	M	Built	Polling faster than 3 s returns `slow_down`; an expired code cannot be approved; a used code returns `already_used` (Playwright journey)	G, C, A, P
FR-051	Device tokens: 15-minute access, 30-day rotating `hgd_` refresh with reuse detection, session version	M	Built	Reusing a rotated device refresh token revokes the token family and is audited	G, A, P
FR-052	Suspend, reactivate, revoke, reassign; revocation effective on next request and within 30 s on the event stream	M	Built	A revoked device's open stream receives `session.ended` within 30 s	C, P

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-053	Device permission ceiling (`DEVICE_PERMISSIONS`)	M	Built	A device paired to an org admin still cannot manage members	P
FR-054	UI profiles with draft, publish, rollback; precedence device, user, role, org; live push	S	Built	Publishing a profile changes the glasses screens without a reload; rollback restores the previous version	C, G, P
FR-055	Ray-Ban Meta camera through DAT (preview stream and full-resolution photo)	M	Partial	Code complete in `GlassesManager.kt` and the WebView bridge. Gap: not validated on physical glasses	A
FR-056	Glasses capture button takes the page photo or starts a scan	S	Partial	Short press emits a capture in `GlassesManager.captureButton`. Gap: not validated on hardware	A
FR-057	The Android app registers the connected glasses as a Hydris device using the pairing endpoints	M	Partial	`Repository.registerGlasses` runs challenge, lookup, approve (first permitted site) and poll, then stores device tokens. Gap: depends on a real DAT device id; not validated on hardware	A, P
FR-058	Log out and unpair from the glasses menu, ending the device's server sessions	M	Built	After `/device/logout` the device's refresh token is rejected and the glasses can be paired again by anyone	G, A, P
FR-059	Glasses heads-up app with 12 screens and a six-key directional input vocabulary, plus a phone layout	M	Built	Every screen is reachable with up, down, left, right, select and back; browser back never leaves the app	G

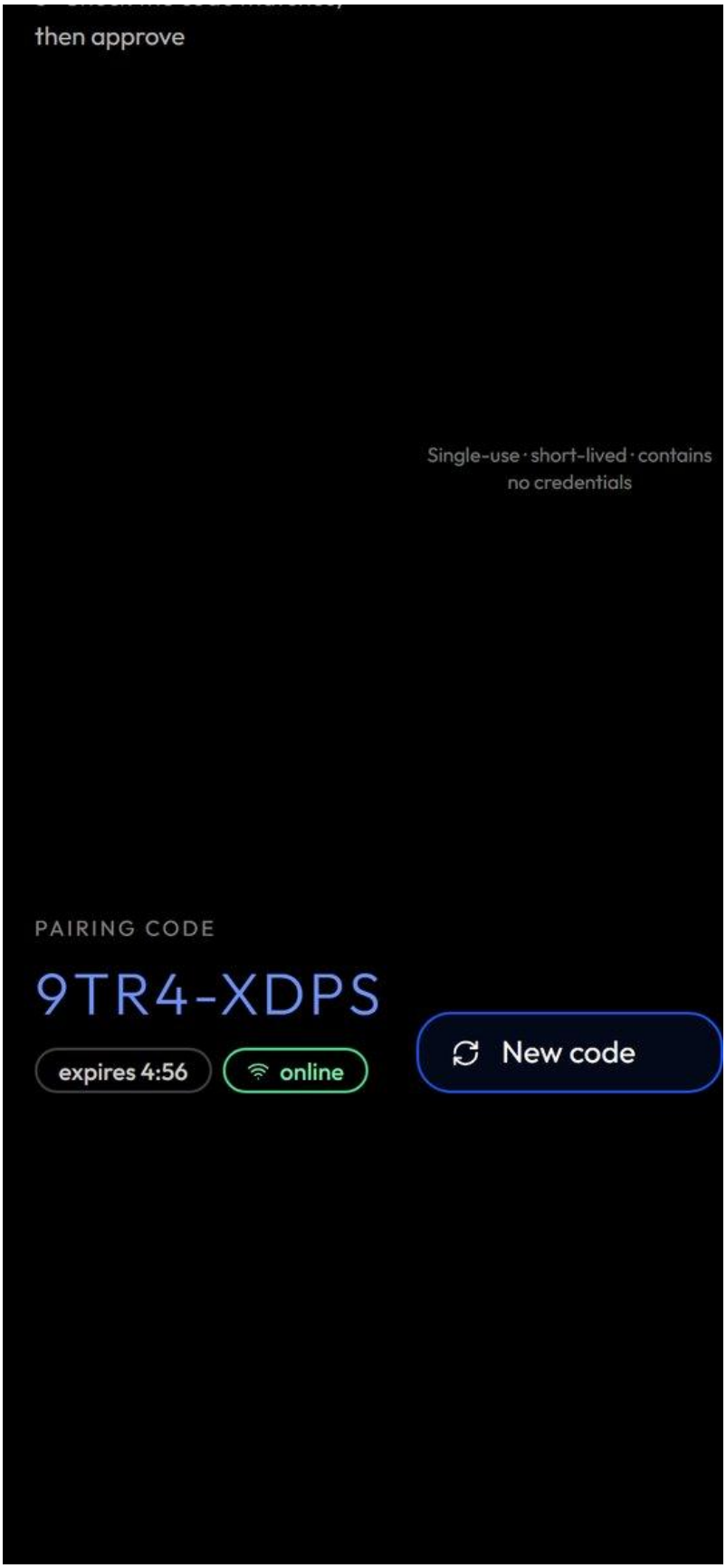


Figure 19. *The glasses app in its phone layout with a pairing code*

7.7 Android app: Hydris Field

The native Android app is the operator's daily driver and the only supported way to reach Ray-Ban Meta glasses for the pilot. It is written in Kotlin with Jetpack Compose, targets Android 12 and later (min SDK 31, matching DAT), and uses the same API and session contract as the web console. It has five tabs (Home, Work, Scan, Hydris, Glasses) plus alarm and sensor detail screens (`apps/android/app/src/main/java/ai/hydris/field/ui/Root.kt`).

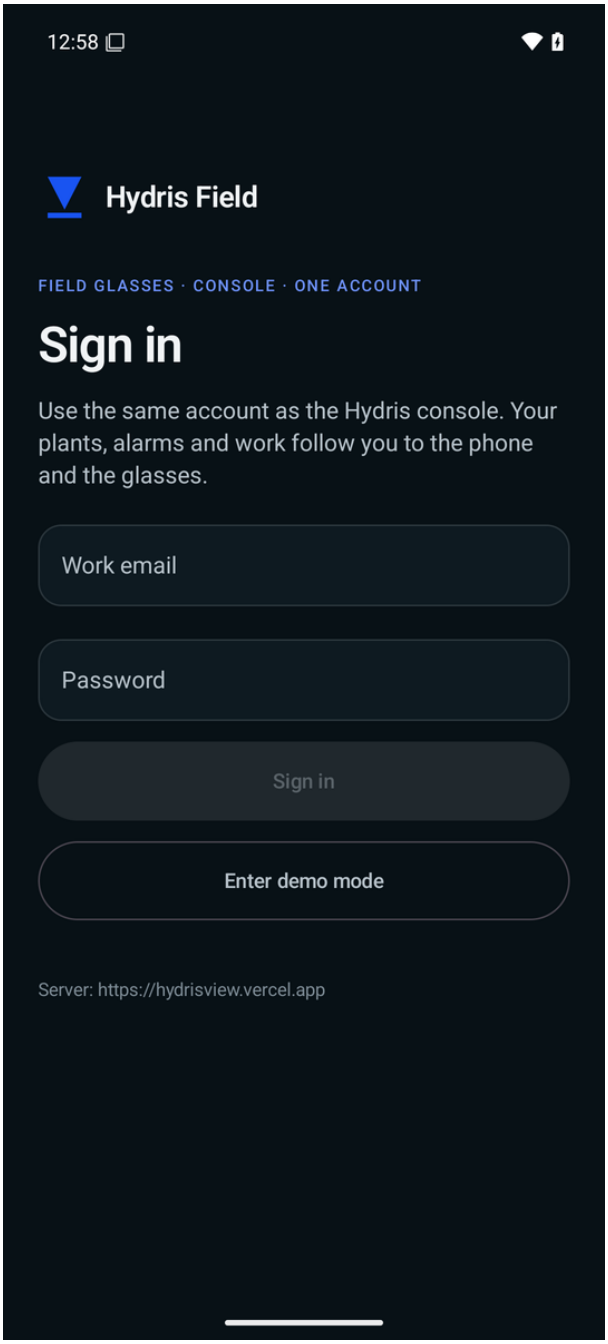


Figure 20. *Sign-in on the Android app*



Figure 21. *Home: greeting, 14-day alarm trend, asset health and live measurements*

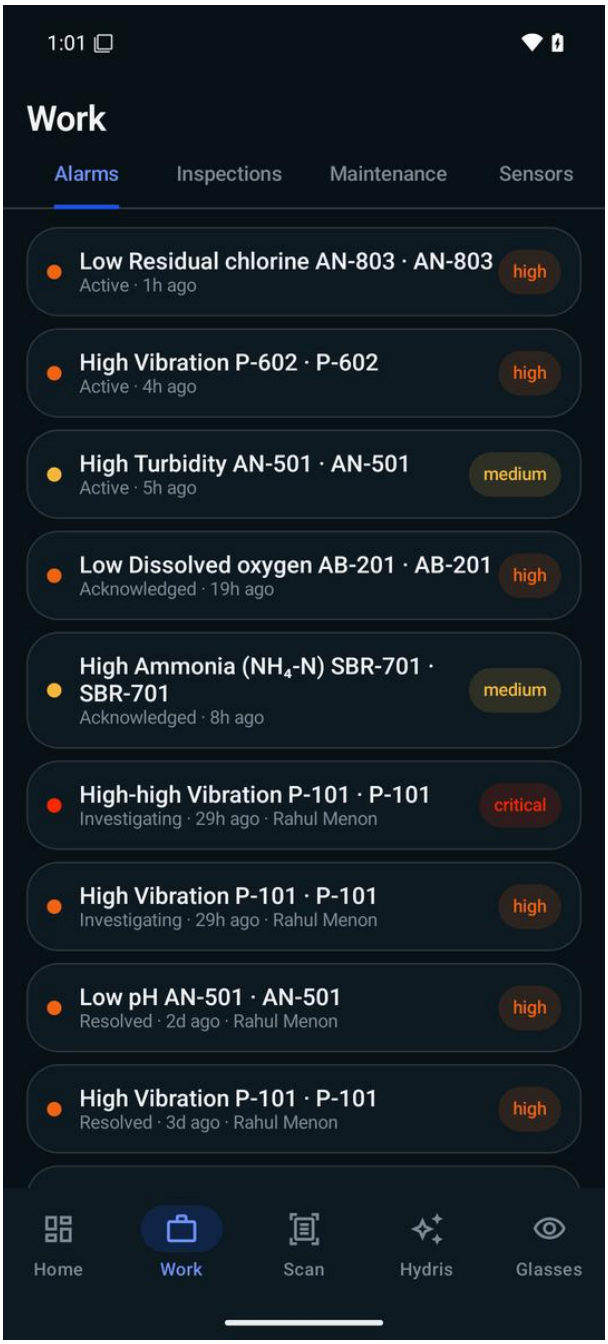


Figure 22. *Work: alarms tab with severity and state*

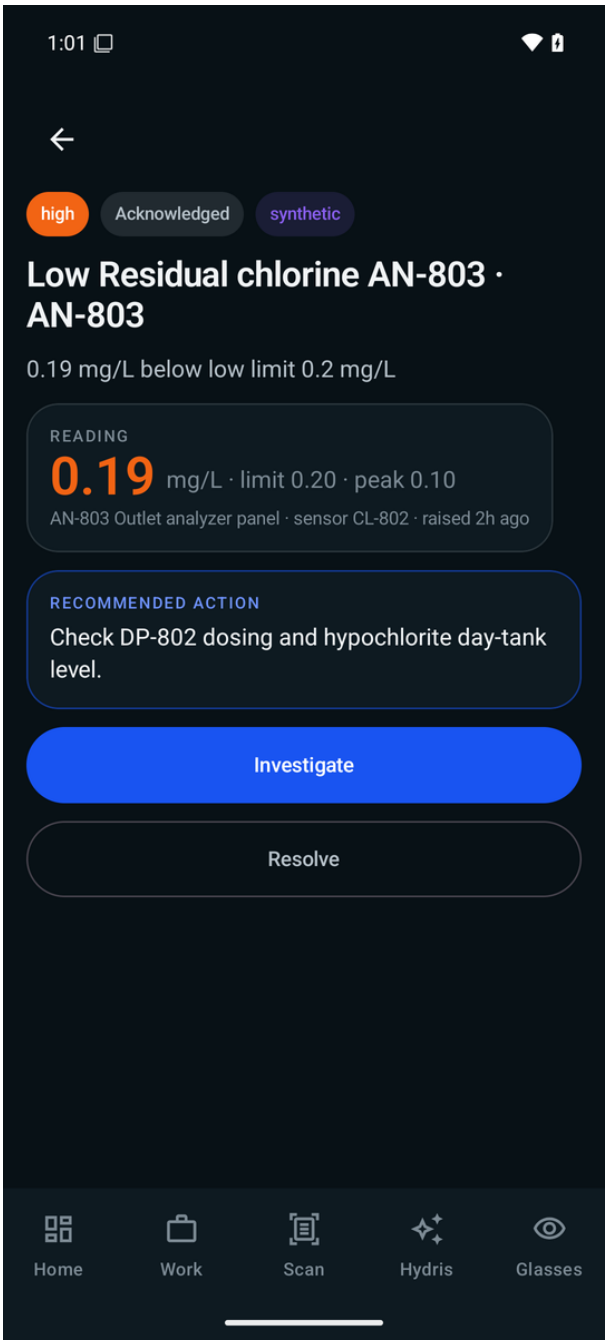


Figure 23. *Alarm detail with reading, limit, peak and the next lifecycle action*

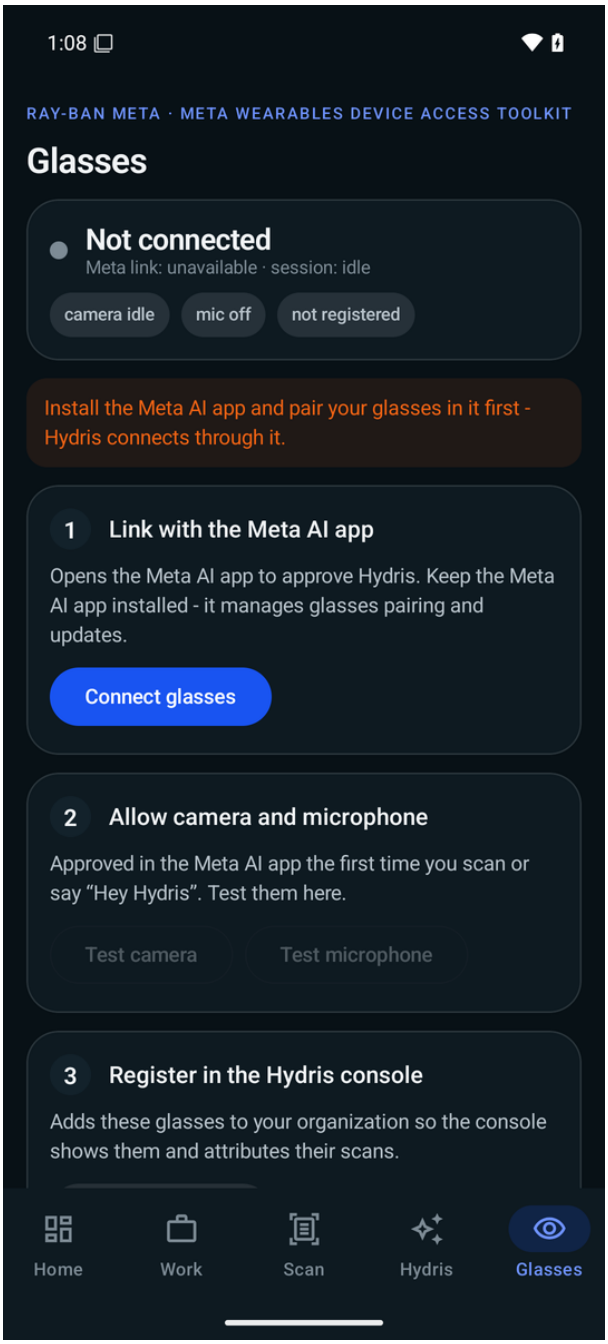


Figure 24. *Glasses: link with the Meta AI app, allow camera and microphone, register in the console*

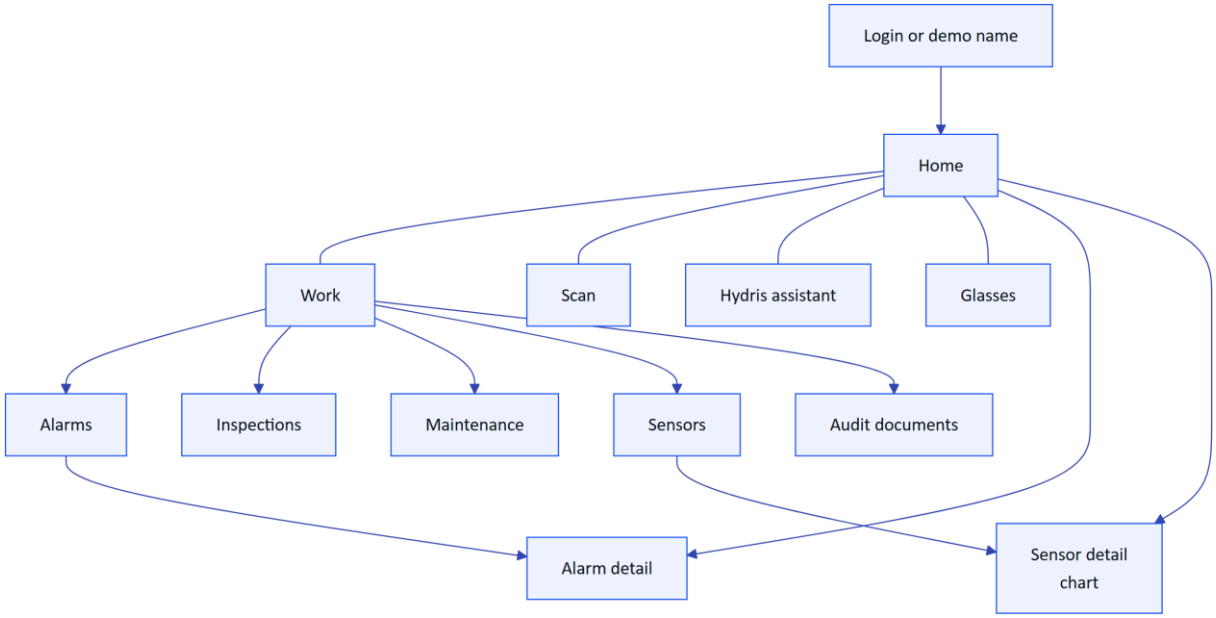


Figure 25. *Android app navigation map*

Table 13. **Android app requirements**

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-090	Sign in with the console account; restore the session on restart from the refresh cookie, stored AES-256-GCM encrypted with a non-exportable Android Keystore key	M	Built	Killing and reopening the app restores the session without a password; no credential is stored in plain text (`SecureStore.kt`)	A
FR-091	Demo mode sign-in with a name	M	Built	Entering a name of 2 or more characters opens a personal demo account	A
FR-092	Home: greeting, 14-day alarm trend, asset health, live measurements with freshness and synthetic labels, recent activity	M	Built	Values marked synthetic in demo; stale values show their state	A
FR-093	Work tabs: alarms, inspections, maintenance, sensors, audit documents	M	Built	Each tab lists only what the user's site scope allows	A
FR-094	Alarm detail with lifecycle actions	M	Partial	Acknowledge, investigate and close work. Gap: Resolve sends no note, so the API returns `note_required`; the app also shows actions without checking `allowedActions`	A, P
FR-095	Sensor detail with a 144-point history chart and min, typical and max	S	Built	Chart covers the history window with typical range from `MEASURAND_DEFS`	A

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-096	Audit scan from the glasses camera, the phone camera or an imported photo, with guided A4 capture, flattening, two ML Kit OCR passes and up to 6 pages	M	Built (phone). Glasses path Partial (FR-055)	A scan of the sample sheet reaches the console in review with values and engine agreement	A, P
FR-097	Realtime alarm notifications: active critical or high alarms raise a system notification while the app runs	M	Built	`alarm.created` on the SSE stream raises a high-priority notification; reconnects resume with `Last-Event-ID`	A, P
FR-098	Offline outbox for actions taken without connectivity (Room-backed, idempotent replay)	S	Planned (GA)	Alarm acknowledgements and checklist ticks made offline are sent once when the network returns	A
FR-099	Complete MFA sign-in in the app	S	Planned (Pilot)	An MFA-enabled operator signs in on the phone with a TOTP code	A, P

RISK

FR-094 is a correctness gap a pilot operator will hit on day one: tapping Resolve in the Android app fails because the API requires a resolution note (FR-031). Fix before pilot: add a note dialog for Resolve and drive the buttons from the API's `allowedActions`.

7.8 Audit document scanning

Audit scanning turns a paper record (inspection checklist, lab report, permit, calibration certificate, maintenance log) into a verified, searchable digital document. The design rule is dual verification: a machine cross-check of every value between independent readings, then a human review of every value and an independent approval by a second person.

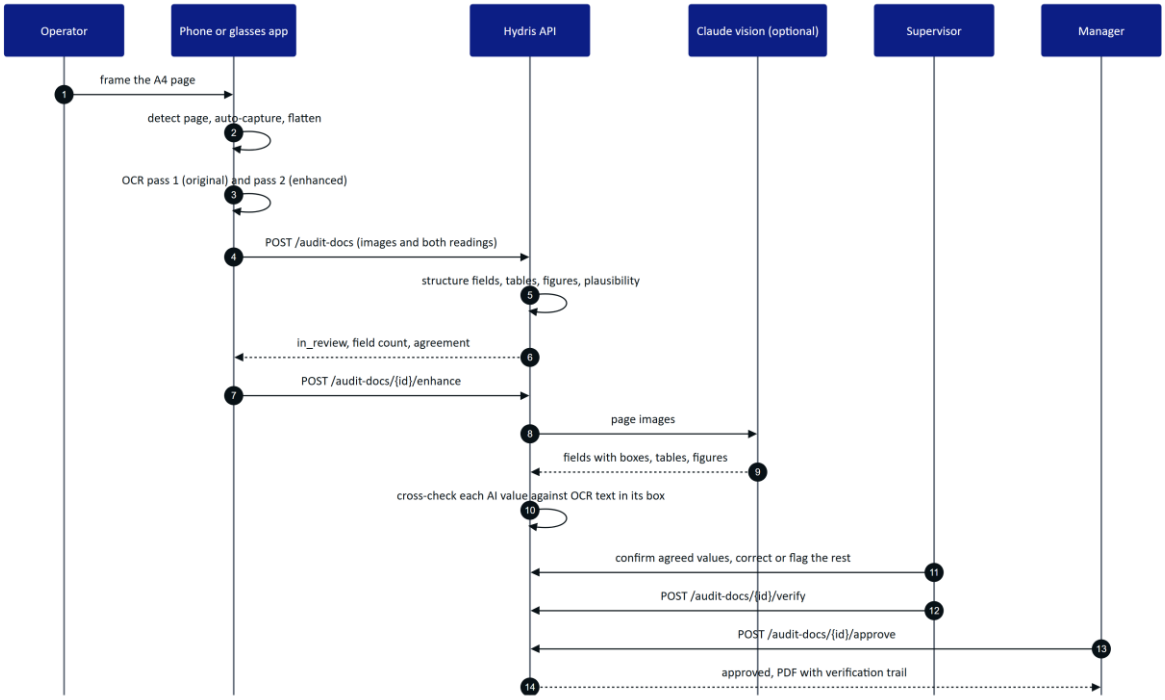


Figure 26. *Scan to verified PDF: capture, machine cross-check, human review, approval*

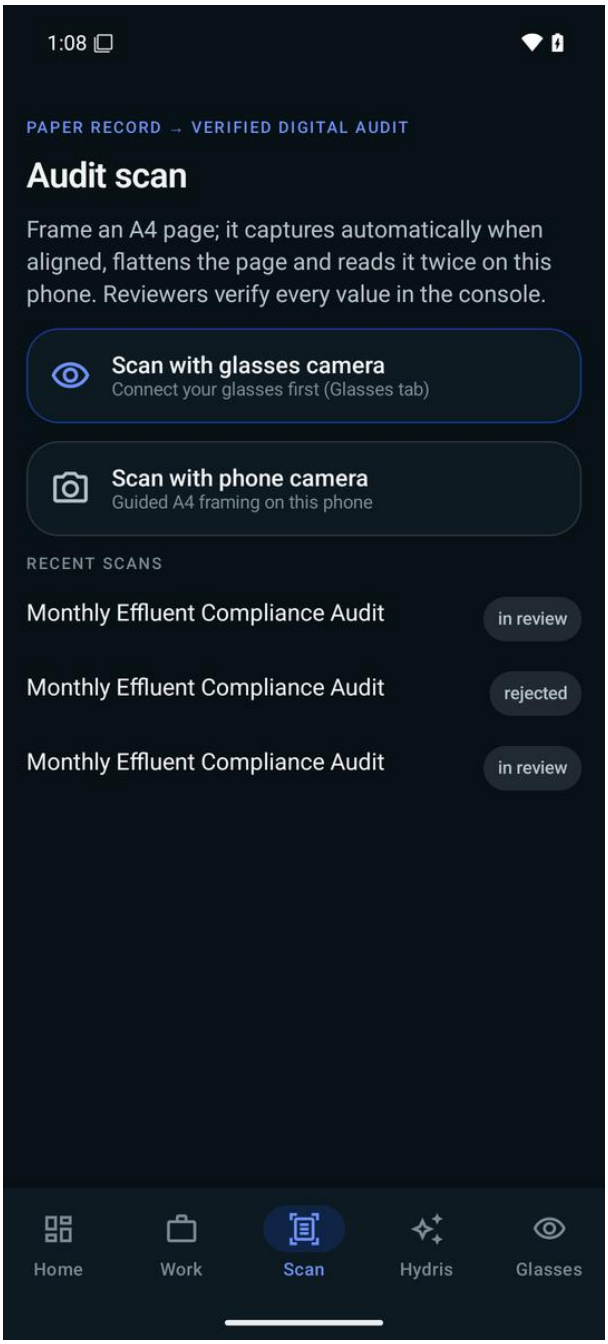


Figure 27. *Scan menu on Android: glasses camera, phone camera or import*

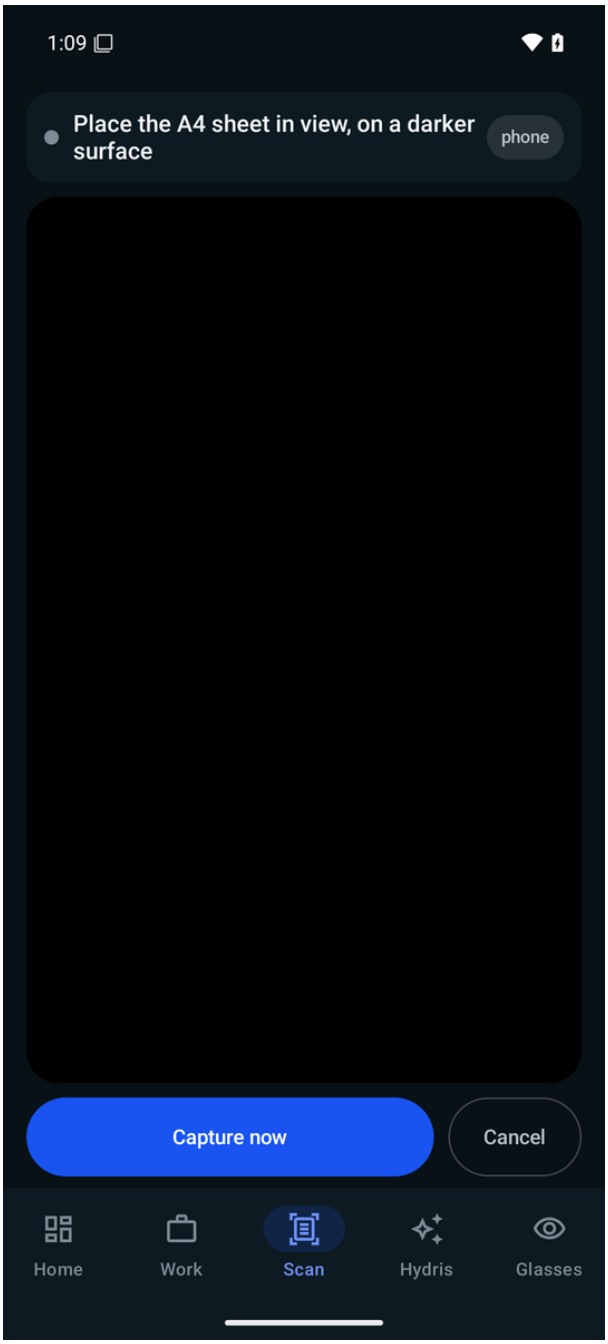


Figure 28. *Guided A4 capture: page detected, aligned and steady before auto-capture*

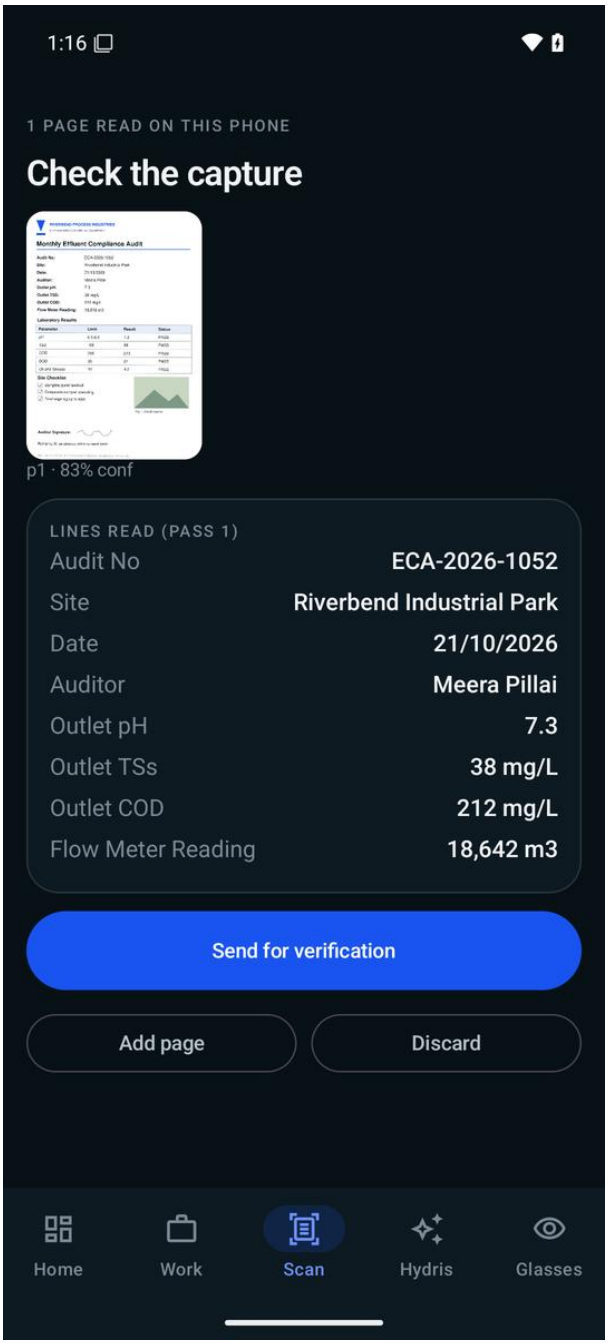


Figure 29. *Review the capture: confidence per page and the label and value pairs found*

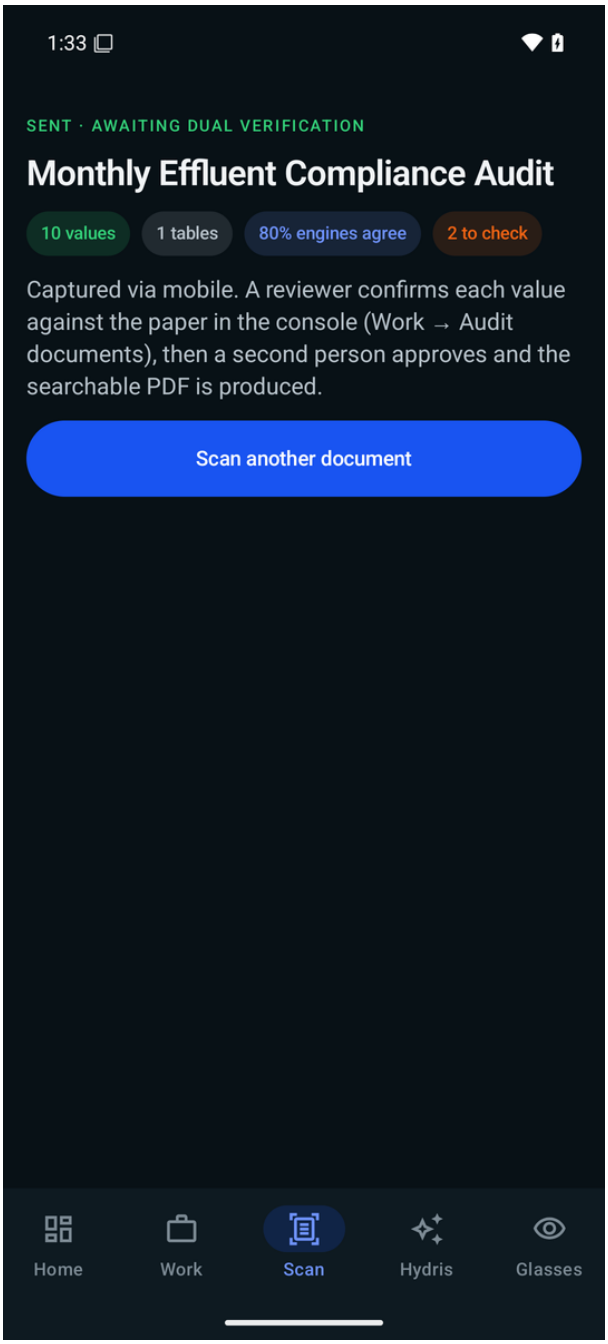


Figure 30. *Sent for verification: the next steps for reviewer and approver*

The document lifecycle is enforced by the API (`apps/api/src/modules/audit-docs.ts`). Every review action is appended to `audit_document_events`; corrections keep the first extracted value; concurrent edits are rejected with a revision check.

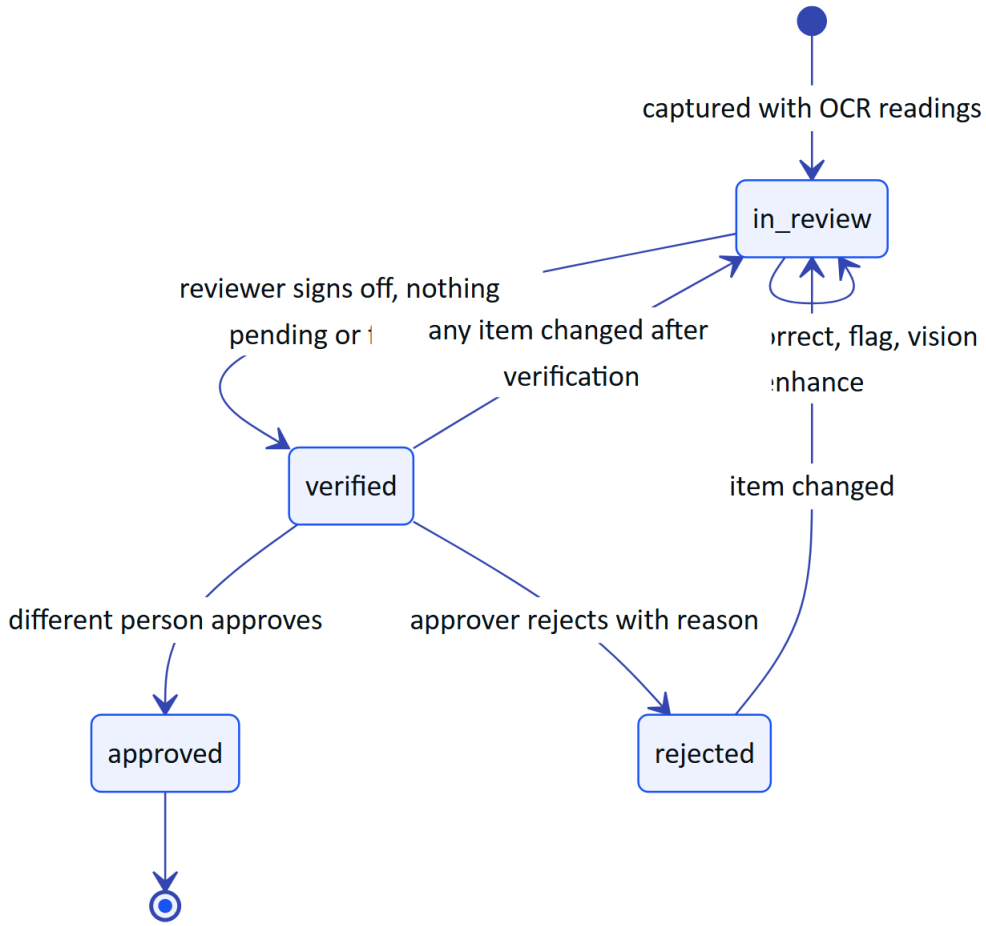


Figure 31. *Audit document lifecycle and the four-eyes rule*

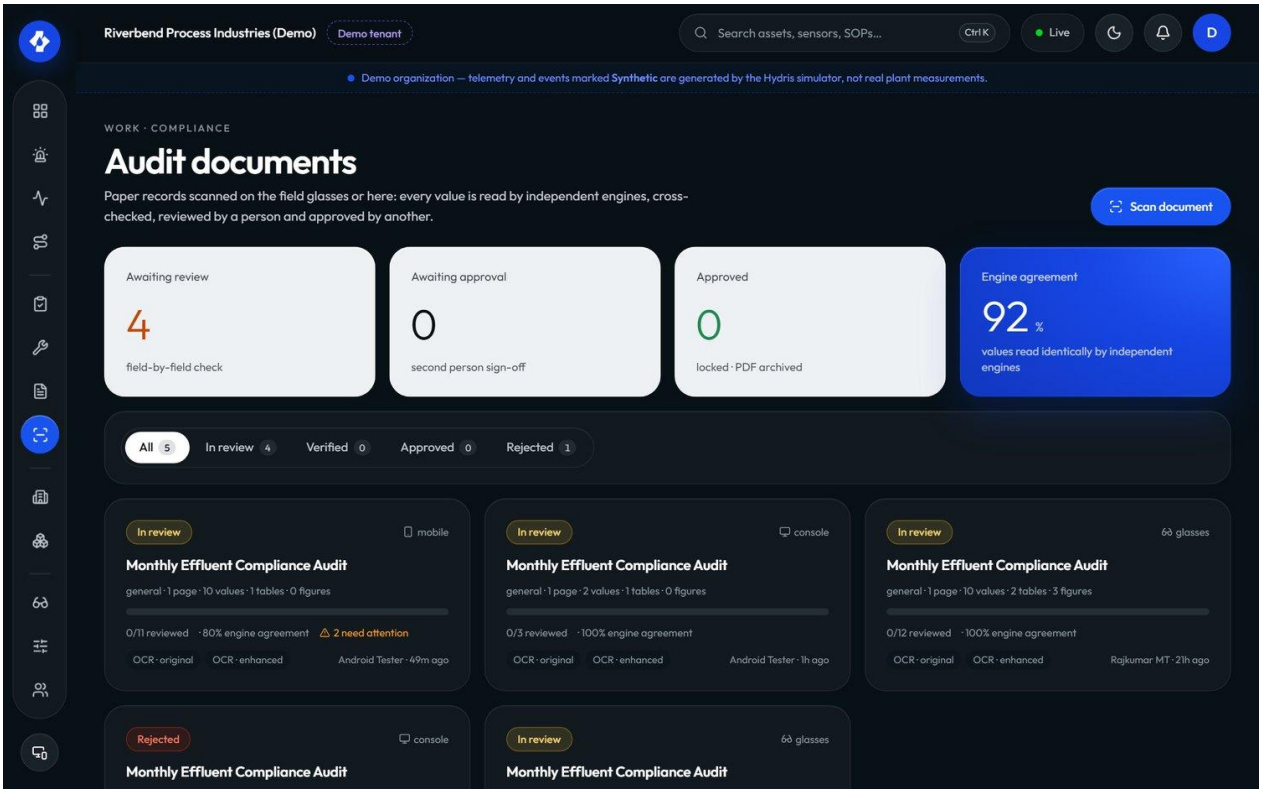


Figure 32. *Audit documents in the console: status, engines, agreement and review progress*

Table 14. Audit document scanning requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-060	Guided A4 capture with live page detection, auto-capture and perspective correction	M	Built (web `docdetect.ts`, Android `DocDetect.kt` with the same thresholds)	Capture fires only when the page is aligned, covers enough of the frame and is sharp	G, A, C
FR-061	Dual-pass on-device OCR (`ocr:original`, `ocr:enhanced`)	M	Built (Tesseract.js on web, ML Kit on Android)	Two readings per page arrive at the API; no page image leaves the device before upload	G, A, C
FR-062	Structuring into typed fields, tables and figures, with plausibility checks (pH 0 to 14, percentages, temperatures)	M	Built (`auditdoc.ts`)	An implausible pH is flagged with a message on the field	P
FR-063	Optional vision engine (Claude) cross-checked against the OCR text inside each value's box	S	Built (needs `ANTHROPIC_API_KEY`)	Each vision value lists the OCR sources that agree or disagree; without a key the OCR result is kept and `aiStatus` is `unavailable`	P
FR-064	Field review: confirm, correct, flag, reset; original value kept; verify requires zero pending and zero flagged items	M	Built	Verify with one flagged item returns `400 flagged_items`; a correction keeps `extracted`	C, P
FR-065	Independent approval (four eyes); self-approval only in demo tenants and recorded as such	M	Built	The verifier approving their own document in a customer org gets 403	C, P
FR-066	PDF with verification trail, searchable original and reconstructed page	M	Built (`core/audit-pdf.ts`)	Text in the PDF original is selectable; the cover lists reviewer, approver and times	C, P
FR-067	Up to 6 pages per document, each image at most 900 KB, JPEG or PNG validated by magic bytes	M	Built	A seventh page or a 1 MB page is refused with a clear message	P
FR-068	Electronic signature with re-authentication and signature meaning (Part 11-style)	S	Planned (Advanced)	Approving asks for the password or MFA again and records the meaning of the signature	C, P
FR-069	Bulk-confirm values where at least two independent engines match with confidence of 0.8 or more	S	Built (`confirm-agreed`)	Only multi-source, matching, confident items change to confirmed; the event lists their IDs	C, P

DECISION

Verification and approval are signed only by a person in the console (`p.kind !== 'user'` is refused). A device can capture and pre-review, but it cannot verify or approve. This keeps the evidentiary steps tied to an authenticated human session, not to a shared pair of glasses.

7.9 Voice assistant and grounded assistance

"Hey Hydris" is a hands-free assistant. The wake phrase is detected on the client with a pattern tolerant of common recognition spellings ("hydras", "hi dris"); the question goes to `POST /assistant/ask`. The API answers with one of two engines:

- **Claude** (`claude-opus-5-5` with tool use) when an Anthropic credential is configured: natural phrasing and multi-step lookups through five read-only tools (`get_readings`, `get_open_alarms`, `get_my_work`, `get_asset`, `search_procedures`), at most five tool rounds.
- **A deterministic rules engine** otherwise, and as the fallback when the model fails: readings, alarms, my work, asset status, procedures and help. It never guesses; anything else gets "I can't answer that from your plant data".

Every tool is scoped to the caller (RLS plus site scope), so the assistant can only say what the caller could see in the console. Answers return `facts` (label, value, unit, time, source) so the client can show the evidence under the spoken answer.

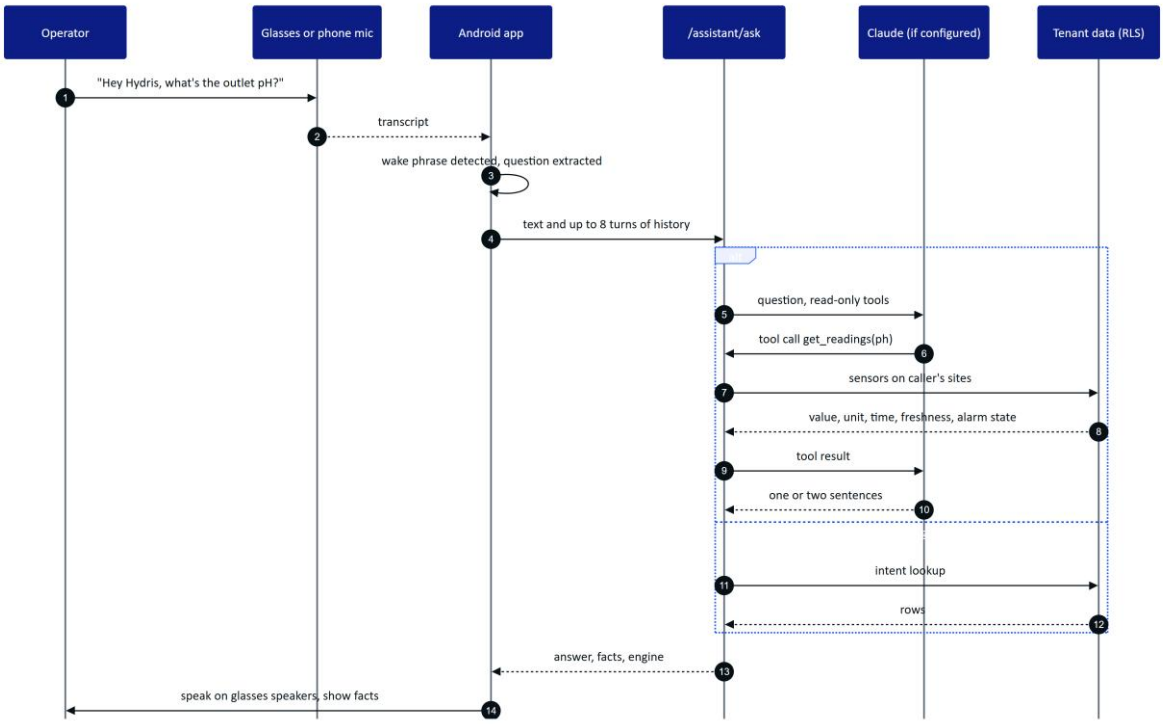


Figure 33. *A spoken question answered from the caller's own plant data*

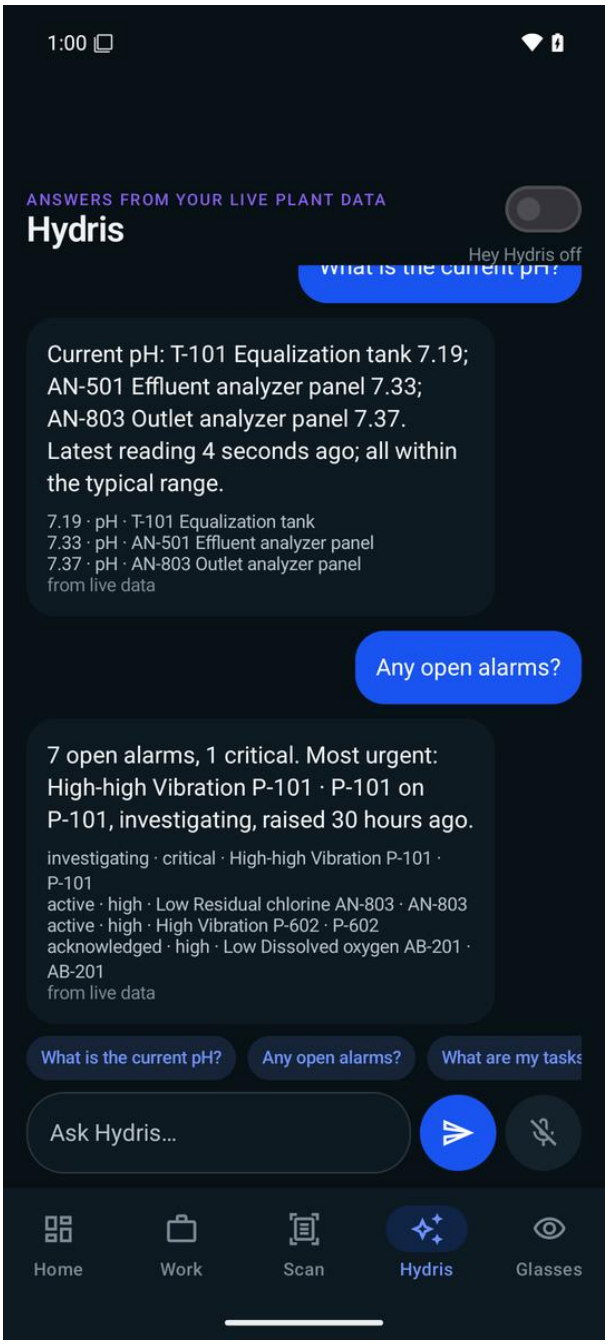


Figure 34. *The Hydris assistant on Android: answers with the facts behind them*

Table 15. Voice and assistance requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-070	Grounded SOP retrieval (`/assist`) with passages from approved documents and a safety notice	M	Built	Responses contain only approved documents and always include the safety notice	G, C, P
FR-071	"Hey Hydris" wake phrase and short spoken answers	M	Built (web Web Speech API; Android SpeechRecognizer and TextToSpeech)	Saying the wake phrase and a question produces a spoken answer; "stop" or "thanks" ends the follow-up window	G, A

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-072	`/assistant/ask` answers only from tenant data through scoped tools; rules fallback never guesses	M	Built	An unknown question returns intent `unknown`; every number in an answer appears in `facts` for rules answers	G, A, P
FR-073	Multi-language voice (Hindi, Tamil, Marathi, Gujarati)	S	Planned (Advanced)	A Hindi question returns a Hindi answer with the same grounding guarantees	G, A, P
FR-074	Glasses microphone through DAT Speech with phone-microphone fallback; answers played through TextToSpeech on the glasses' open-ear speakers	M	Partial	Phone path built in `VoiceEngine.kt`. Gap: glasses microphone path not validated on hardware	A
FR-075	Answers show the facts behind them with source, and label synthetic data	M	Built	A demo answer's facts show "synthetic demo data"	G, A, P
FR-076	Follow-up questions with conversation history (up to 8 turns accepted by the API)	S	Built	"And the inlet?" after a pH question is answered for the inlet	G, A, P
FR-077	"Hey Meta, start Hydris" voice invocation opens the assistant	C	Partial (WebView bridge only; DAT feature is experimental)	Voice invocation intent opens the assistant screen	A
FR-078	The assistant has no write tools and cannot change any record or equipment state	M	Built	The tool list contains only read tools; the system prompt forbids operating equipment	P

7.10 Documents and SOPs

Table 16. Document and SOP requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-110	Document library (SOP, manual, datasheet, safety notice, troubleshooting) with draft, approved and archived states, versions, and site, asset or asset-type scope	M	Built	Only `document.manage` holders approve; versions increment on change	C, P
FR-111	Only approved documents reach the glasses, the assistant and `/assist`	M	Built	A draft SOP never appears in a glasses search or assistant answer	G, A, P
FR-112	Alarm rules link a procedure document shown with the alarm	M	Built	Alarm detail shows the linked SOP title and opens it	C, G, P
FR-113	SOPs read on the glasses are cached for offline reading	S	Built (`glasses/data.ts` cache)	A previously opened SOP opens with no network	G
FR-114	Upload PDF manuals and SOPs with text extraction for retrieval	C	Planned (GA)	An uploaded PDF becomes searchable by `/assist` after approval	C, P

7.11 Insights, reports and platform

Table 17. Platform requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-080	Realtime server-sent events with `Last-Event-ID` replay; memory or database mode	M	Built	Reconnecting with the last ID replays missed events; DB mode works across serverless instances	C, G, A, P
FR-081	Reports (inspections, alarms, maintenance, devices, operators) in JSON and CSV	M	Built	Alarm report includes median minutes to acknowledge; inspection report includes on-time rate and glasses share	C, P
FR-082	Append-only audit trail	M	Built	No API path updates or deletes audit events	C, P
FR-083	Demo mode: zero-configuration synthetic tenant, named guests, cross-instance pairing	M	Built	A deployment with no environment variables serves a working demo	All
FR-084	Deterministic simulation scenarios, demo tenants only	M	Built	Ten scenarios run; starting one in a customer org is refused	C, P
FR-085	In-app notifications	M	Built	Notifications appear in realtime and can be marked read	C, G, A, P
FR-086	Email delivery (SMTP, SES or Resend) for `mail_outbox`	M	Planned (Hardening)	Invitation and reset emails arrive; failures retry and are visible to admins	P
FR-087	Audit log page with filters	M	Built	Auditor filters by action and entity and sees actor, time and IP	C, P
FR-088	Global search across assets, sensors, documents and work	S	Built (`/search`)	Typing an asset tag finds the asset and its open work	C, P
FR-089	Overview dashboard with KPIs, open alarms, health and live measurements	M	Built (`/overview`)	Console overview and Android home load from the same endpoint	C, A, P

Table 18. Insights and compliance requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-120	Compliance summary per discharge point: daily minimum, maximum and average against consent limits, with data availability	S	Planned (Pilot)	A manager exports a monthly summary for the final outlet with gaps and exceedances marked	C, P
FR-121	Scheduled report delivery by email	C	Planned (GA)	A weekly alarm report arrives every Monday (depends on FR-086)	P
FR-122	Shift handover summary generated from the shift's alarms, work and readings, grounded and editable before sending	C	Planned (GA)	The outgoing operator reviews a summary whose every number links to a record	A, C, P

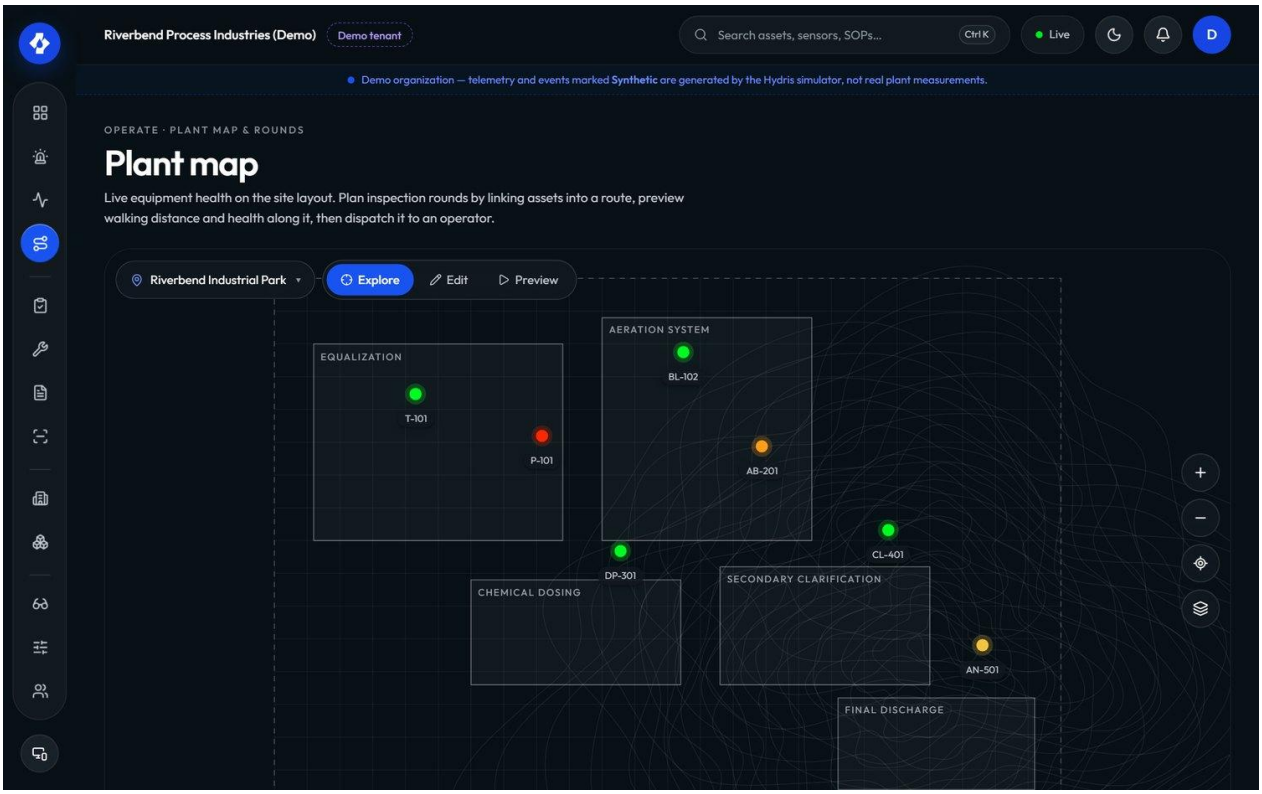


Figure 35. *Plant map and rounds in the console*

7.12 Website, demo and public

Table 19. Website, demo and public requirements

ID	Requirement	Pri	Status	Acceptance criteria	Surface
FR-100	Marketing website with seven pages and a privacy page	M	Built	Pages render on phone and desktop, light and dark	W
FR-101	Contact and demo-request form `POST /public/inquiries`, rate limited to 5 per 10 minutes per client	M	Built	The sixth submission within 10 minutes is refused	W, P
FR-102	Live glasses demo from the website (`/glasses?demo=1&scenario=...`), fully client-side, and the browser simulator	M	Built	The demo runs with no API calls and shows scenario events	W, G
FR-103	Named demo guests: each guest is an `org_admin` in the demo tenant with personal copies of scheduled inspections and open tasks	M	Built	Two guests acting at once do not see each other's personal work changes	C, A, P

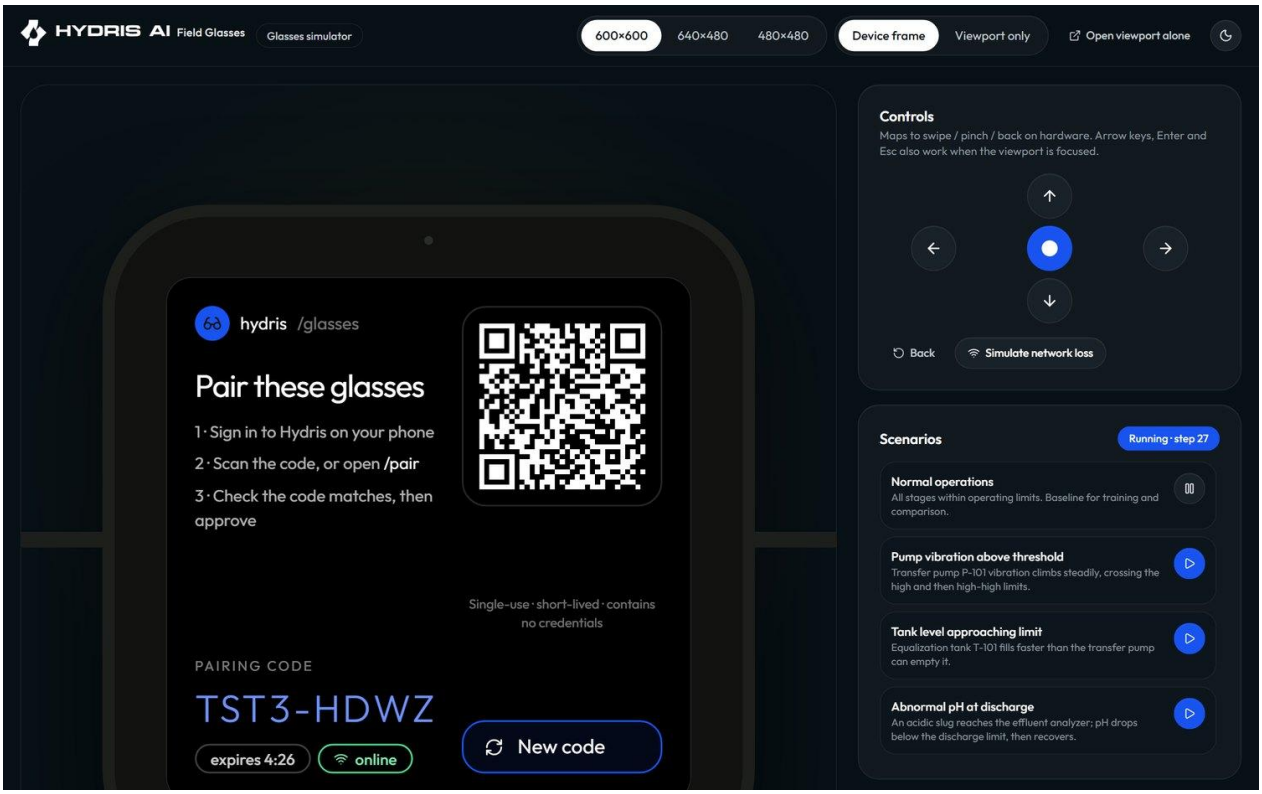


Figure 36. *The browser glasses simulator used for demos and testing*

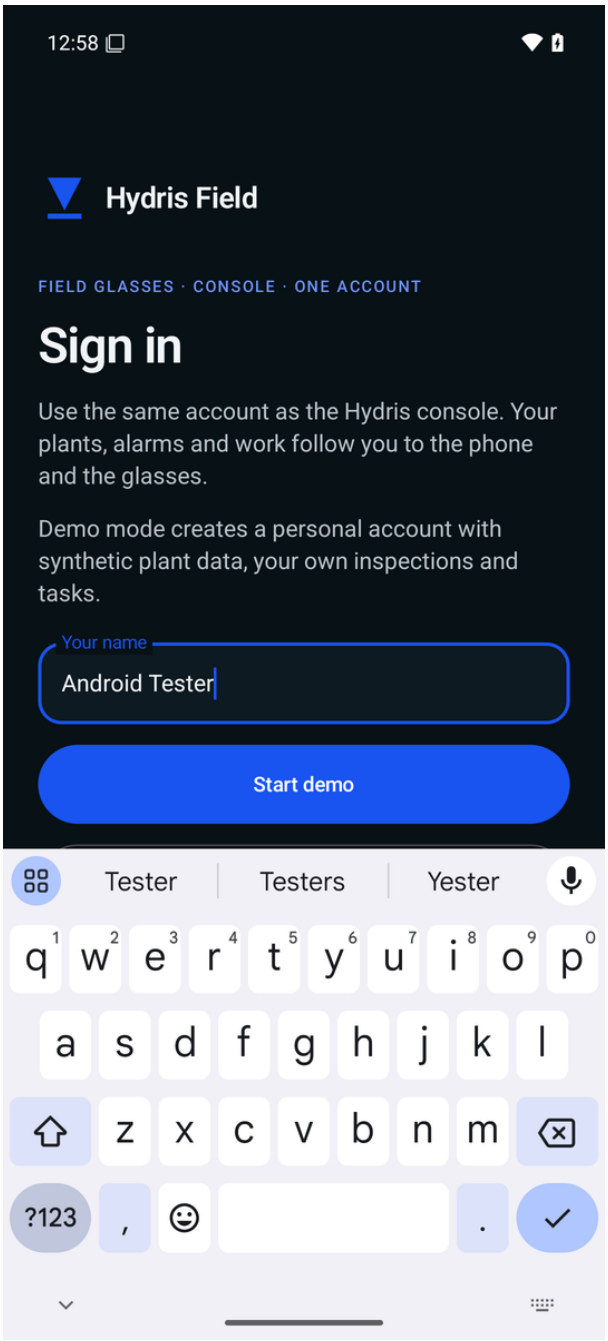


Figure 37. *Demo mode on Android: enter your name to get a personal demo account*

7.13 Requirement status overview

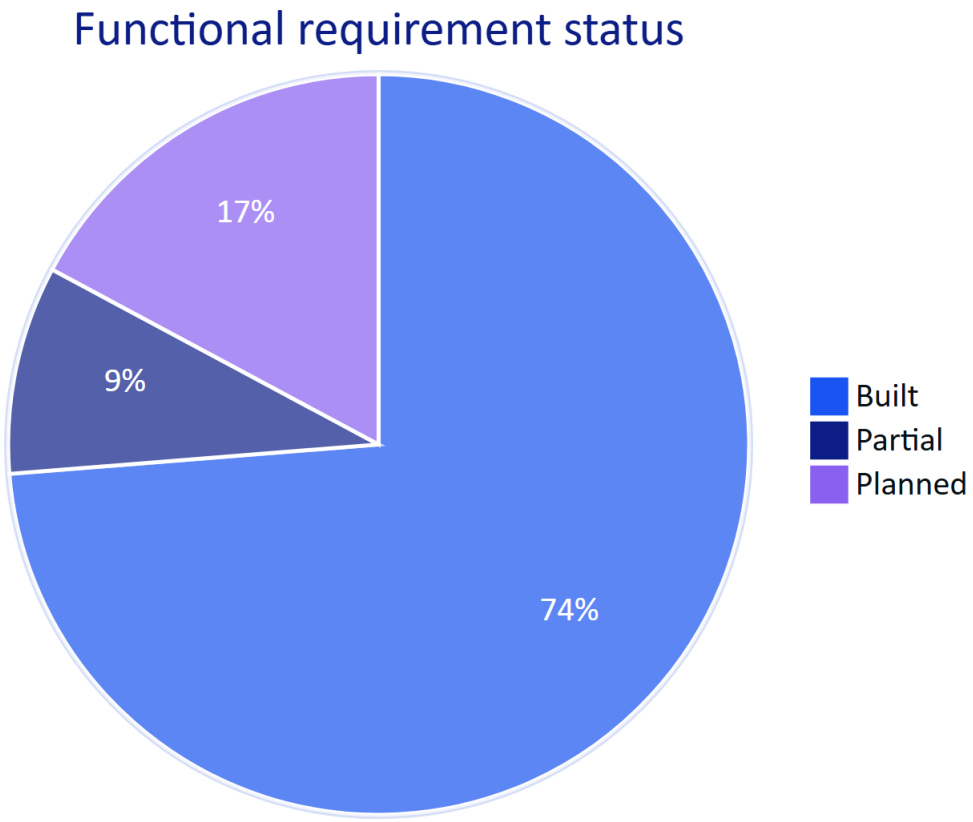


Figure 38. *Status of all functional requirements in this edition*

8 Non-functional requirements

Non-functional requirements apply across surfaces. IDs are new in this edition (version 1.0 listed the same areas without IDs). Values are targets unless a measurement is named.

8.1 Performance and responsiveness

Table 20. Performance requirements

ID	Requirement	Pri	Status	Target and verification
NFR-001	API read latency	M	Partial (not yet measured)	Target p95 under 300 ms for reads in-region; verify with load tests before pilot
NFR-002	Realtime alarm delivery from creation to client	M	Built (design); not measured	Target 2 s or less. Memory mode is immediate; DB mode polls every 1.5 s (`realtime.ts`)
NFR-003	Voice answer time from end of speech to start of spoken answer	M	Partial (not measured)	Targets: 3 s with the rules engine, 8 s with the model engine
NFR-004	OCR time per page on a mid-range phone	S	Partial (not measured)	Target 15 s or less per page for both passes
NFR-005	Glasses home screen ready after wake	S	Planned (measure)	Target 2 s from cached context; fresh data within 5 s

8.2 Availability, scalability and operations

Table 21. Availability and operations requirements

ID	Requirement	Pri	Status	Target and verification
NFR-006	Service availability	M	Planned (measure)	Targets: 99.5 % monthly in pilot, 99.9 % at GA, measured by external uptime checks
NFR-007	Horizontal scale of realtime	M	Built	`REALTIME_MODE=db` supports serverless and multiple instances; memory mode is single-instance only
NFR-008	Scheduled jobs (stale rules, pairing expiry, event pruning) run at a cadence suitable for real data	M	Partial	Long-running server: 30 s, 60 s and hourly timers. Vercel: daily cron only; fix in Hardening
NFR-009	Backups and restore	M	Planned	Daily backups, point-in-time recovery on managed Postgres, restore tested each quarter
NFR-010	Continuous integration on every change	M	Planned (Hardening)	Typecheck, unit, API and e2e tests on every pull request; no `.github` workflows exist today

8.3 Security and privacy

Table 22. Security and privacy requirements

ID	Requirement	Pri	Status	Target and verification
NFR-011	Tenant isolation enforced in the database	M	Built	RLS on every tenant table; requests run as `hydris_app` without BYPASSRLS; API tests cover cross-tenant access
NFR-012	Short-lived credentials	M	Built	15-minute access tokens (12 h in demo mode only); rotating refresh tokens with reuse detection
NFR-013	Rate limiting	M	Built	Global 600 requests per minute per client; sign-in 10 per minute; pairing endpoints 10 to 40 per minute; public form 5 per 10 minutes
NFR-014	Secrets at rest	M	Built	MFA secrets encrypted with `DATA_KEY`; tokens stored as SHA-256 hashes; Android credentials in Keystore-backed AES-256-GCM
NFR-015	Camera only on explicit intent; no continuous recording	M	Built	Camera streams only while a scan screen is open (`startPreview` and `stopPreview`); no background capture
NFR-016	Images stored only for audit documents and attachments	M	Built	No other path persists camera frames
NFR-017	Data residency in India for Indian customers	S	Planned (GA)	Production database, storage and model inference in an Indian region, documented per tenant
NFR-018	Independent penetration test before GA	M	Planned (GA)	No open high or critical findings at launch

> **WARNING**
>
> The glasses web app keeps its device refresh token in browser `localStorage` (`apps/web/src/glasses/link.ts`). That is acceptable for the simulator and demo, not for production glasses. The native Android app already stores device tokens in the Keystore-backed `SecureStore`, which is one reason the native app, not the WebView bridge, is the pilot path.

8.4 Accessibility, glanceability and hands-free use

Table 23. Accessibility and hands-free requirements

ID	Requirement	Pri	Status	Target and verification
NFR-019	Glasses text scale per profile	M	Built	Scales 0.9, 1, 1.15 and 1.3 (`UI_TEXT_SCALES`), clamped to 0.85 to 1.4
NFR-020	Additive-display friendly design	M	Built (assumption to verify on hardware)	Pure black renders as transparent; 600 by 600 viewport assumption (`GLASSES_DISPLAY`)
NFR-021	Every glasses action reachable with six directional keys or voice	M	Built	Keyboard, swipe and simulator controls map to up, down, left, right, select, back

ID	Requirement	Pri	Status	Target and verification
NFR-022	Spoken answers are short	M	Built	One or two sentences (assistant system prompt); units spoken in words
NFR-023	Console and Android meet WCAG 2.2 AA	S	Partial (not audited)	Automated and manual accessibility audit before GA; contrast in light and dark themes
NFR-024	Large touch targets for gloved use on Android	S	Built	Primary buttons are 52 to 54 dp tall in the scan, login and alarm screens

8.5 Offline, battery and device constraints

Table 24. Offline and device requirements

ID	Requirement	Pri	Status	Target and verification
NFR-025	Inspection responses survive connectivity loss on the glasses	M	Built	Queue plus idempotent replay; Playwright journey and `glasses_network_loss` scenario
NFR-026	Last context and SOPs readable offline on the glasses	S	Built	Cached with timestamp; shown as cached, never as live
NFR-027	Android offline outbox	S	Planned (GA)	See FR-098
NFR-028	Battery: a 4-hour shift on Ray-Ban Meta with up to 20 scans	S	Planned (measure)	Camera streams only on the scan screen; Android preview frames throttled to at most one per 100 ms; speech restarted only while enabled
NFR-029	Android 12 or later (min SDK 31)	M	Built	Matches DAT requirements

8.6 Data integrity and maintainability

Table 25. Data integrity and maintainability requirements

ID	Requirement	Pri	Status	Target and verification
NFR-030	Provenance on every reading; synthetic data impossible in customer tenants	M	Built	DB check `telemetry_synthetic_source_ck`; ingest refuses simulator data for non-demo orgs
NFR-031	Optimistic concurrency on review records	M	Built	Audit document edits carry a revision; stale writes get `409 stale`
NFR-032	Shared domain logic between server and clients	M	Built	State machines, permissions, freshness and voice parsing live in `packages/domain`; Android ports keep identical thresholds
NFR-033	Automated test coverage of security-critical paths	M	Built	67 tests passing (26 domain, 41 API) on 22 Oct 2026; 4 Playwright journeys
NFR-034	Observability	S	Planned (Pilot)	Structured logs, error tracking and uptime alerts with an on-call runbook

9 Data and AI requirements

9.1 Data principles

1. **Tenant data stays in the tenant.** All AI calls are made with data the caller is already allowed to see, fetched through the same RLS-scoped queries as the console.
2. **The model never sees more than it needs.** The assistant receives tool results for the question asked, not a dump of the database. Vision extraction receives only the page images of the document being processed.
3. **Nothing the model says becomes a record without a person.** Vision-extracted values enter the review as `pending`; a reviewer confirms each one and a second person approves.
4. **Every AI output is attributable.** Audit documents record which engines produced each value (`engines`, `sources`); assistant responses say which engine answered (`engine: claude-opus-5-5` or `hydris-rules`).

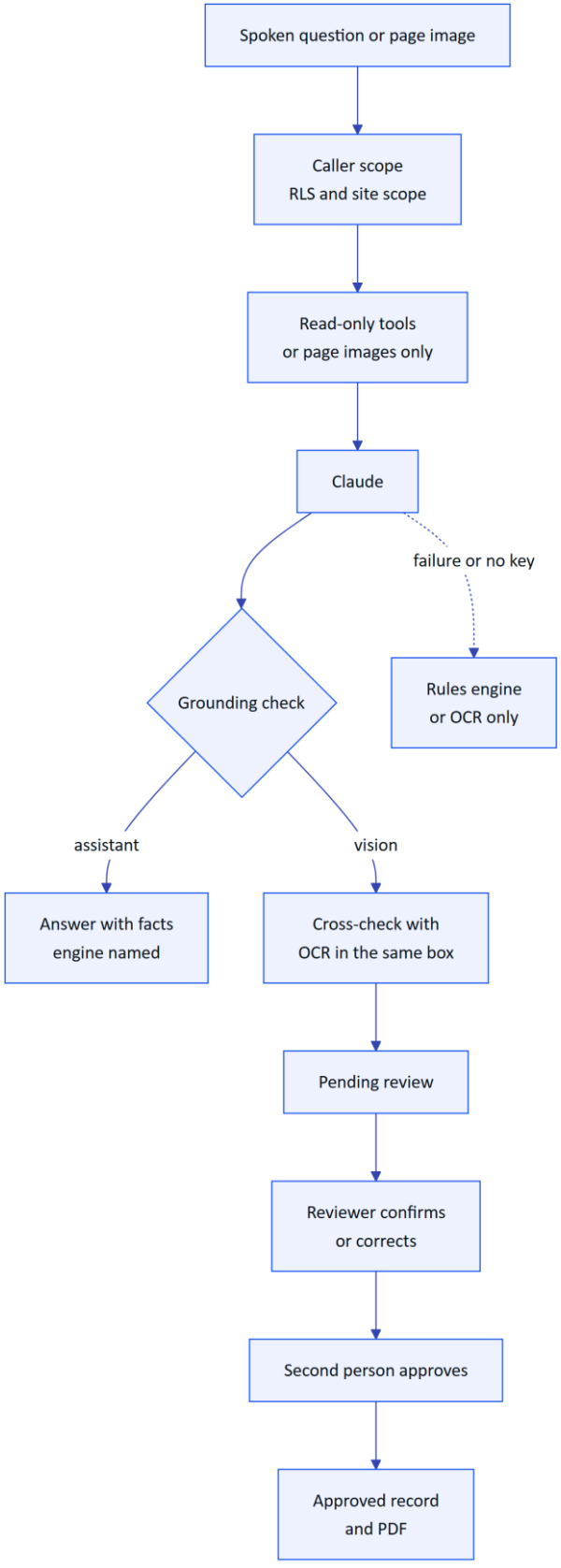


Figure 39. *Guardrails between the language model and the plant record*

9.2 Grounding requirements

Table 26. Data and AI requirements

ID	Requirement	Pri	Status	Acceptance criteria
NFR-035	The assistant uses only values returned by tools in the same conversation	M	Built (prompt and tool design)	Evaluation set: zero answers containing a number absent from tool results (target)
NFR-036	The assistant states the age of a reading and says plainly when it is stale, missing, out of range or in alarm	M	Built	Rules answers include "measured N minutes ago"; stale readings add "not current, check the sensor"
NFR-037	Graceful degradation without the model	M	Built	Any model error falls back to the rules engine; vision failure keeps OCR results with `aiStatus = failed` and the reason
NFR-038	Vision transcribes faithfully: no correcting, normalizing or inferring illegible values	M	Built (system prompt)	Illegible areas appear in warnings with low confidence
NFR-039	Human-in-the-loop for every AI-extracted value	M	Built	No document reaches `verified` with pending or flagged items
NFR-040	Model choice and data handling documented per tenant, including region	S	Planned (GA)	Tenant settings show which model provider and region process its data

9.3 Accuracy targets

Table 27. AI and OCR quality targets (targets, not measurements)

Measure	Target for pilot	How it will be measured
Fields where independent engines agree (`agreement = match`)	90 % or more on typed forms	Agreement score per document in the console; pilot sample of 200 pages
Fields a reviewer has to correct	under 5 %	Ratio of `corrected` to all reviewed fields
Vision false confidence (value confirmed by engine agreement but wrong on paper)	under 0.5 %	Blind re-check of 100 approved documents
Assistant answers with facts (not "can't answer")	85 % or more of real operator questions	Intent logging on `/assistant/ask` (engine, intent)
Assistant answers with a number not in its facts	0	Automated check on logged answers and facts

> **IMPORTANT**
>
> The assistant and vision engine run on `claude-opus-5-5` when `ANTHROPIC_API_KEY` is set (`voice-assistant.ts`, `core/vision-extract.ts`). Before a customer pilot the tenant must agree to that processing, and the deployment must record where inference runs. Until then, pilots can run with the rules engine and OCR only, which are fully on Hydris infrastructure and the device.

10 Success metrics and KPIs

10.1 North star

Verified field records per active operator per week: the number of inspections submitted, alarms acknowledged within target, and audit documents approved, divided by the number of operators who used Hydris that week. It rises only when operators use the product in the field and the records it produces are trusted.

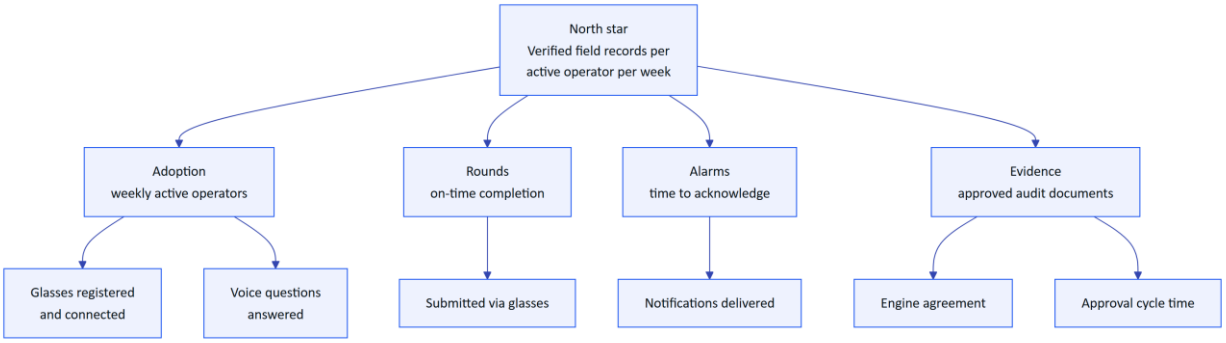


Figure 40. *KPI tree: the north star and the input metrics that drive it*

10.2 Metrics and targets

Table 28. KPIs with definitions, sources and pilot targets

Area	Metric	Definition and source	Pilot target
Adoption	Weekly active operators	Operators with at least one record in a week divided by licensed operators	70 % or more
Rounds	On-time completion	`onTimeRate` in the inspections report	95 % or more
Rounds	Time per inspection	Median start to submit, compared with the paper baseline	30 % lower than paper
Rounds	Share submitted via glasses	`viaGlasses` in the inspections report	50 % or more
Alarms	Time to acknowledge	`medianMinutesToAcknowledge` in the alarms report	under 5 minutes for critical and high
Alarms	Acknowledged but unresolved backlog	Count of alarms in acknowledged or investigating older than 24 h	Visible daily; trending down
Evidence	Engine agreement	Average agreement score of approved documents	90 % or more
Evidence	Reviewer edits	Corrected fields divided by reviewed fields	under 5 %
Evidence	Approval cycle	Capture to approval, median	under 24 hours
Voice	Answered with facts	Share of questions with intent other than `unknown`	85 % or more
Trust	Hallucinated numbers	Numbers in answers missing from facts	0
Security	Cross-tenant incidents	Confirmed data exposure across tenants	0
Security	Device revocation time	Revoke to `session.ended` on the stream	30 s or less

> **NOTE**
> Most adoption and round metrics already have a data source in `/reports/:kind` (`apps/api/src/modules/insights.ts`). The voice and agreement metrics need lightweight logging that is Planned for the pilot (part of NFR-034).

11 Release plan and milestones

11.1 What has shipped

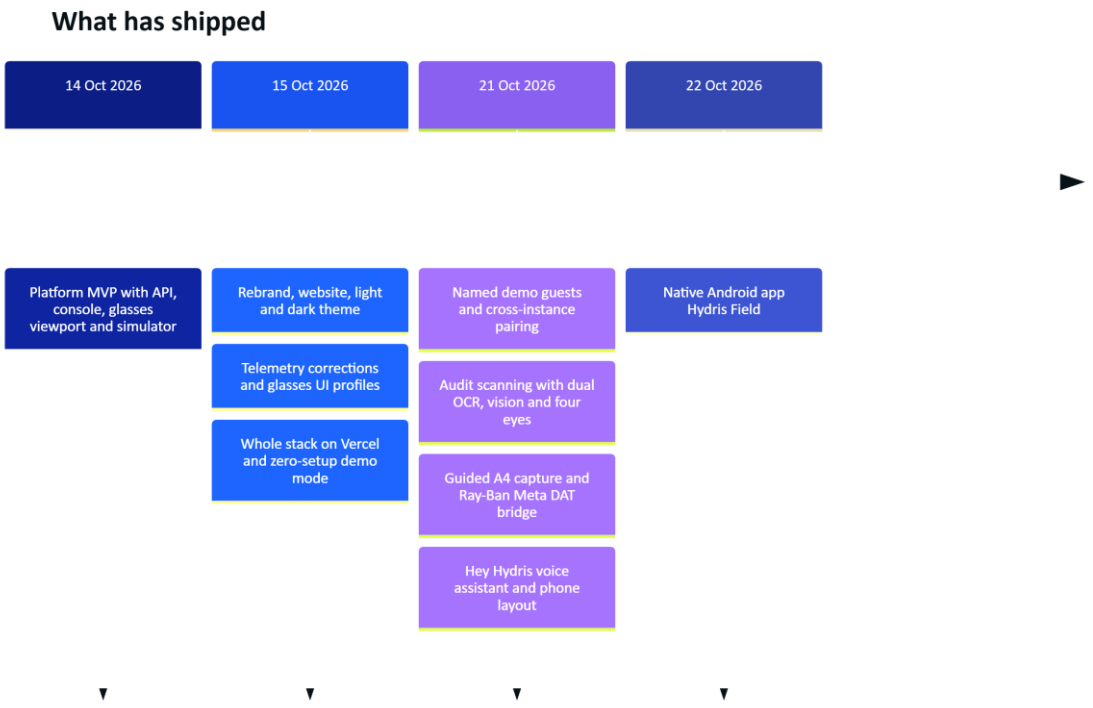


Figure 41. *Build history on main (git log, October 2026)*

11.2 Phases

Table 29. Release phases, scope and exit criteria

Phase	When	Scope	Exit criteria
0. MVP	Done (to October 2026)	API, console, glasses app, simulator, pairing, UI profiles, audit scanning, voice, demo mode, native Android app in code	Live demo; 67 tests passing
1. Hardening	Q4 2026	Email transport (FR-086), CI (NFR-010), job cadence on serverless (FR-035, NFR-008), Android fixes (FR-094, FR-099), audit page images to object storage, observability baseline	All Must requirements for pilot are Built or have an agreed workaround
2. Hardware validation	Q4 2026 to Q1 2027	Ray-Ban Meta on the test matrix: camera, capture button, speech, registration (FR-055, FR-056, FR-057, FR-074); battery measurement (NFR-028)	Every glasses requirement moves from Partial to Built or is descoped with a phone fallback
3. Pilot	Q1 2027	One or two Indian plants (ETP or STP), 10 to 30 operators, gateway ingestion, compliance summary (FR-120), escalation (FR-037), Android inspections (FR-044)	KPI baselines captured; pilot targets met or explained; customer sign-off

Phase	When	Scope	Exit criteria
4. GA	Q2 2027	FCM push (FR-034), Android offline outbox (FR-098), SSO (FR-011), India-region production (NFR-017), pen test (NFR-018), recurring schedules (FR-047), Play or MDM distribution	Store or MDM release approved; security review passed
5. Advanced	H2 2027 onwards	MQTT and OPC UA (FR-027), SCADA connectors (FR-028), multi-language voice (FR-073), e-signatures (FR-068), native glasses display, predictive maintenance, LIMS, iOS	Driven by customer demand

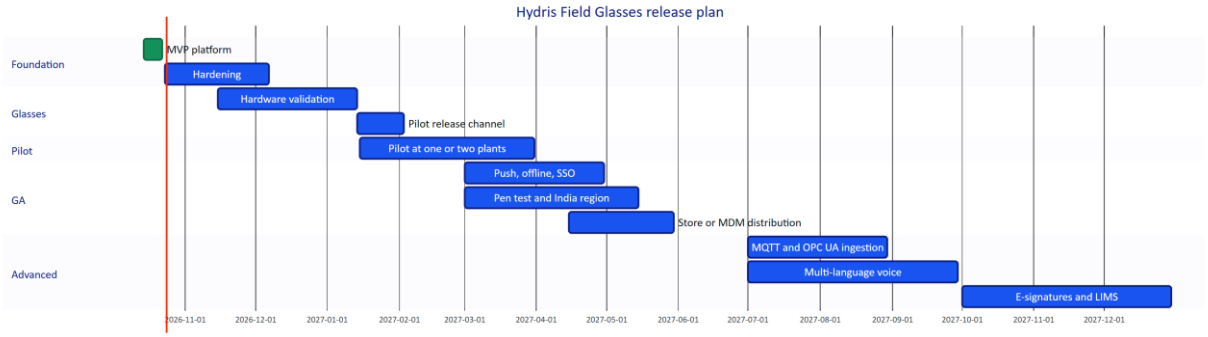


Figure 42. *Release plan (indicative dates)*

11.3 Prioritisation

Planned and Partial work was scored for value to the pilot customer and engineering effort by the product team. The chart is a planning aid, not a measurement.

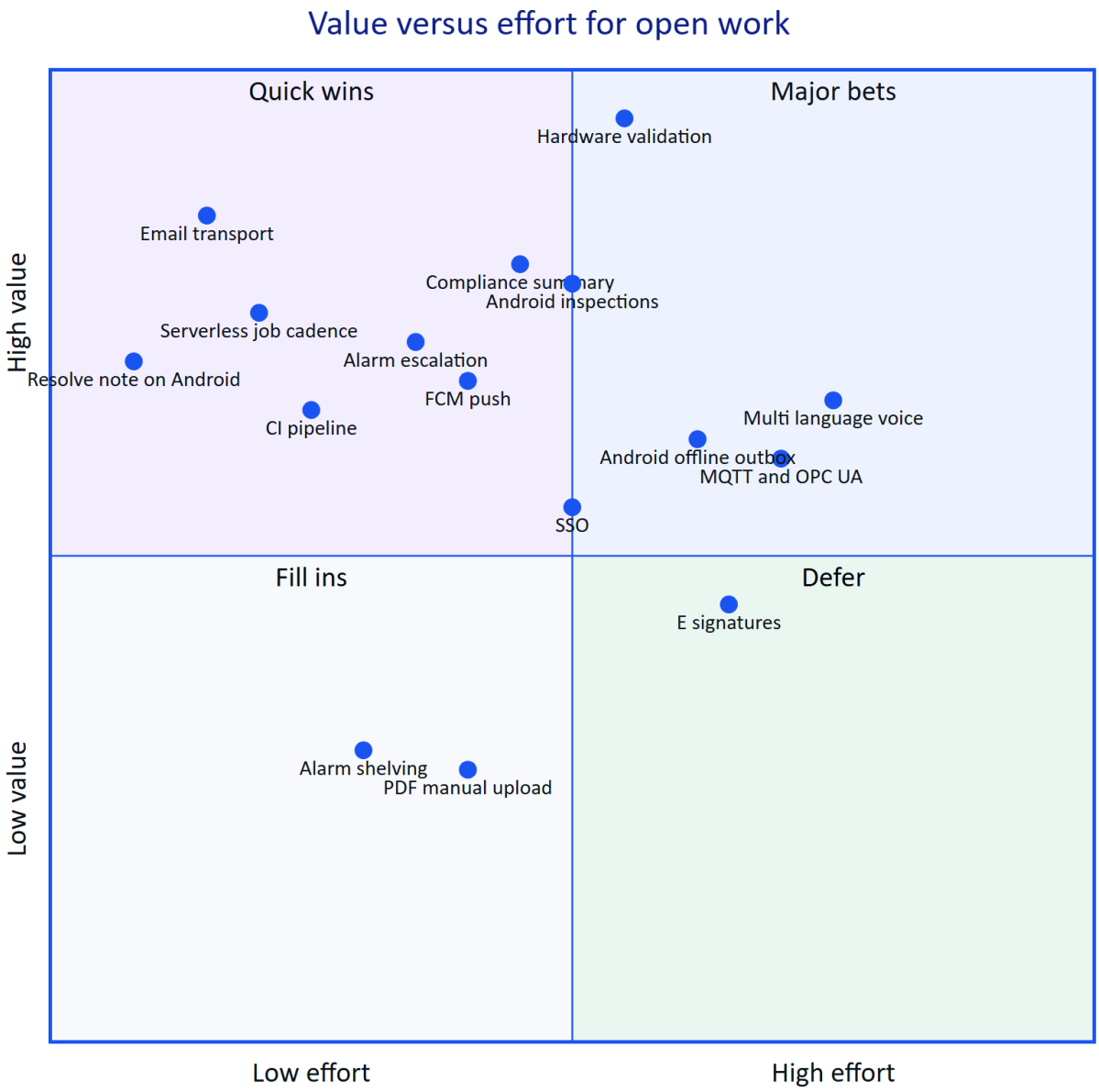


Figure 43. *Planned and partial work by value and effort (product team estimate)*

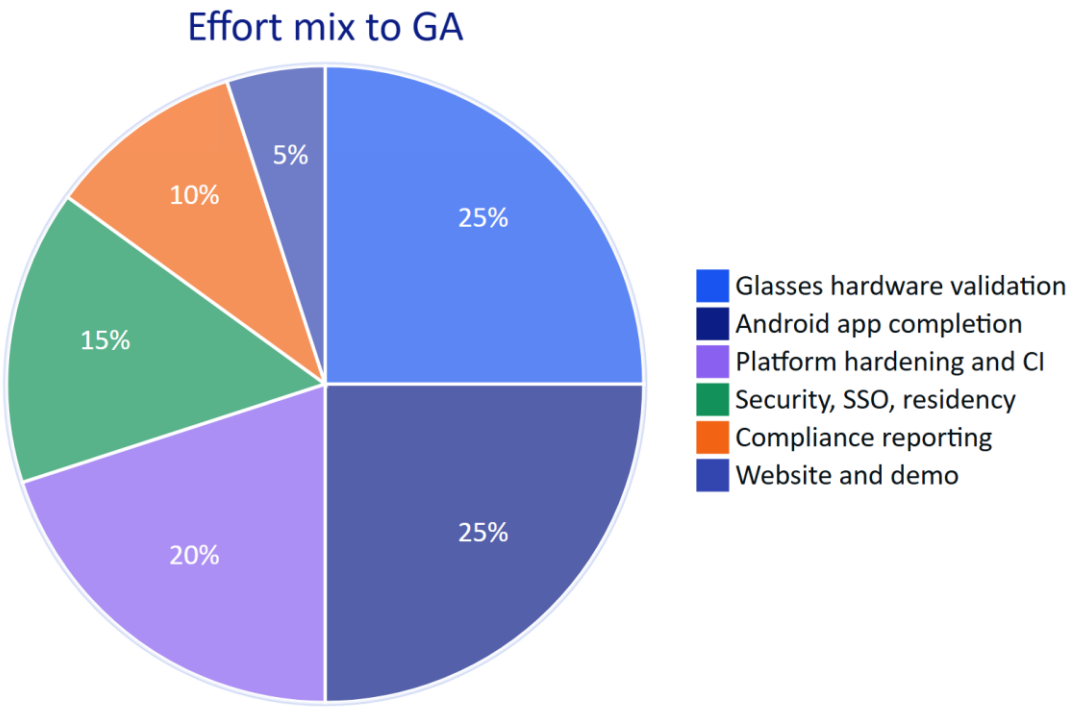


Figure 44. *Where the remaining effort goes before GA (product team estimate, share of engineering weeks)*

DECISION

The pilot runs on the native Android app, not the WebView bridge. The native app keeps device tokens in the Android Keystore, uses ML Kit for faster on-device OCR, and talks to DAT directly. The bridge remains useful for demonstrations of the web glasses app on hardware.

12 Dependencies

Table 30. External dependencies, what Hydris uses them for, and the fallback

Dependency	Used for	Owner	Risk if unavailable	Fallback
Meta AI app (`com.facebook.stella`)	Glasses pairing, DAT registration, camera and microphone permissions, firmware	Meta	Glasses features unavailable	Phone camera and phone microphone paths work without it
Meta Wearables DAT 1.0.1 and Developer Mode	Camera stream, photo, capture button, speech	Meta	Cannot reach the glasses	Same as above
Wearables Developer Center registration	Production app ID and client token; publishing beyond developer mode	Meta	Cannot distribute beyond test devices	Developer Mode for the pilot; apply early
Vercel	Hosting web, API route handlers, cron, Blob storage	Vercel	Outage of the hosted service	Container deployment (Docker, Render blueprint) is supported
Postgres (Neon or managed)	Primary database with RLS	Provider	Outage	Point-in-time recovery; PGlite for local and demo
Anthropic API	Assistant model and vision extraction	Anthropic	Degraded answers, no vision	Rules engine and OCR-only extraction are built
Google ML Kit text recognition	On-device OCR in the Android app	Google	Lower OCR quality	Tesseract.js path in the web app
Email provider (Planned)	Invitations, resets, scheduled reports	To be chosen	Invitations and resets do not arrive	Admin copies links from `mail_outbox`

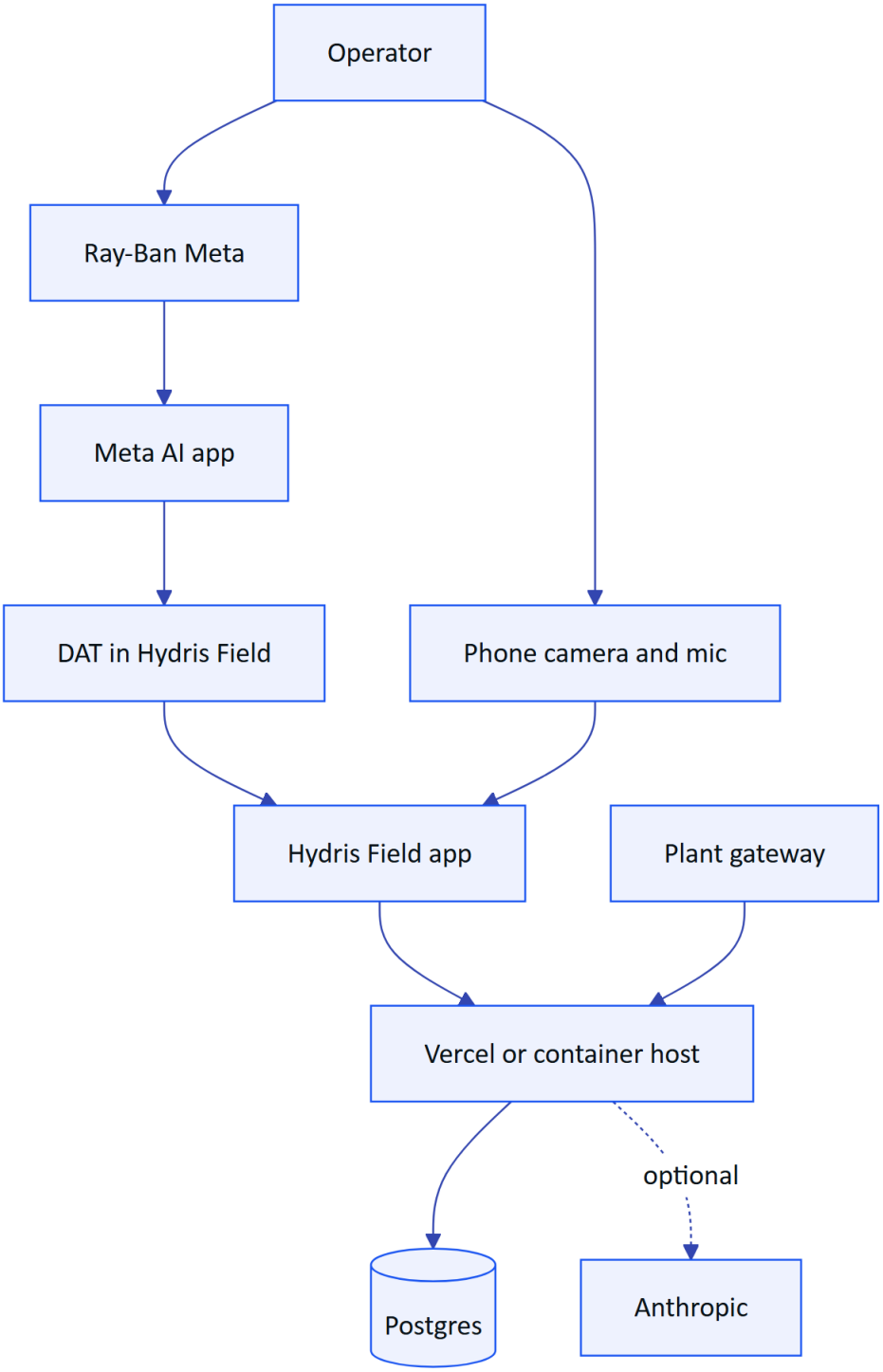


Figure 45. *Dependency chain from operator to plant data*

13 Risks and mitigations

Table 31. Product risks, impact, likelihood and mitigations

#	Risk	Impact	Likelihood	Mitigation	Owner
R1	Meta DAT publishing remains gated; store submissions not open when we need them	Cannot distribute glasses features at scale	Medium	Pilot in Developer Mode and Developer Center release channels; phone paths ship independently; apply early	Product
R2	Meta AI app dependency (pairing, registration, permissions, firmware) changes behaviour	Support burden, version drift	Medium	Onboarding checklist; the Glasses screen checks for the Meta AI app and shows link state and a connection log	Mobile
R3	Glasses paths fail on real hardware	Scan and voice on glasses unusable	Medium	Phase 2 hardware matrix before pilot; phone fallback on every glasses feature	Mobile, QA
R4	Email not wired	Invitations and resets do not reach users	High until fixed	FR-086 in Hardening; interim admin copy of links	Platform
R5	Serverless cron runs once a day	Stale-data alarms late on Vercel deployments with real data	High on Vercel Hobby	Move to per-minute cron (Pro plan) or a container host; NFR-008	Platform
R6	No CI	Regressions reach the live demo	Medium	NFR-010 in Hardening	Platform
R7	Android Resolve fails without a note	Operators cannot close the loop on the phone	High	FR-094 fix in Hardening	Mobile
R8	LLM availability, cost or data-residency objections	Degraded answers; pilot blocked by policy	Medium	Rules engine and OCR-only modes; region-pinned provider option; NFR-040	Product
R9	Camera glasses not accepted in a customer's security zones; bystander privacy	Feature disabled on site	Medium	Camera only on intent; UI profile can disable the Scan screen; site camera policy agreed in the pilot contract	Product
R10	Regulators or auditors do not accept Hydris PDFs as primary records	Paper remains the record	Medium	Keep the searchable original scan inside every PDF; FR-068 e-signatures; early conversations with the SPCB through the pilot customer	Product
R11	Device refresh token in `localStorage` on the web glasses app	Token theft on a compromised device	Low for pilot (native app)	Native app with Keystore storage is the pilot path	Security
R12	Audit page images stored in the database	Database growth and cost	Medium at scale	Move page images to object storage in Hardening	Platform

RISK

The single largest schedule risk is R3 combined with R1: everything that makes Hydris unique on glasses (camera scanning, capture button, glasses microphone) is code-complete but unproven on hardware, and distribution beyond developer mode is in Meta's hands. The plan protects the pilot by making every glasses feature work on the phone first.

14 Open questions

Table 32. Open questions, owner and when they must be answered

#	Question	Owner	Needed by
Q1	Is India among Meta's supported countries for full DAT capabilities and Developer Center access? Written confirmation needed	Product	Before Phase 2
Q2	When do DAT 1.0 store submissions open, and what review does an unlisted enterprise app need?	Product	Before GA planning
Q3	Will pilot customers allow camera glasses inside their plants, and in which zones?	Customer success	Pilot contract
Q4	Which regulators accept a Hydris-generated PDF as the primary record rather than a scanned copy?	Product, pilot customer	Pilot exit
Q5	Is a separate iOS app needed for pilot or GA customers?	Product	GA planning
Q6	Must LLM inference run in India for Indian customers, and through which provider?	Security	Before enabling the model for a customer
Q7	Should acknowledgement on the glasses count as acknowledgement by the assigned operator, or by the device, for compliance reports?	Product, compliance	Pilot
Q8	What consent-limit data model does FR-120 need (per parameter, per discharge point, per consent version)?	Product	Pilot
Q9	Which email provider meets deliverability and residency needs (SES Mumbai, Resend, other)?	Platform	Hardening
Q10	Should the operator handover summary (FR-122) be generated by the model or by deterministic templates?	Product	GA planning

15 Glossary

Table 33. Terms used in this document

Term	Meaning
ETP	Effluent treatment plant: treats industrial wastewater before discharge
STP	Sewage treatment plant: treats domestic wastewater
WTP	Water treatment plant: produces process or drinking water
CETP	Common effluent treatment plant: shared ETP serving an industrial cluster
CPCB / SPCB	Central and State Pollution Control Boards in India
OCEMS	Online continuous effluent and emission monitoring system
Consent to operate	SPCB permission that sets a plant's discharge limits
Measurand	A quantity measured by a sensor (pH, dissolved oxygen, COD, and so on)
Freshness	Live, delayed, stale or no data, derived from a sensor's expected interval
Deadband	Hysteresis band that stops an alarm chattering around its threshold
On-delay	Time a breach must persist before an alarm is raised
RLS	Row-level security in Postgres, used for tenant isolation
SSE	Server-sent events, the realtime channel from API to clients
DAT	Meta Wearables Device Access Toolkit for Ray-Ban and Oakley Meta glasses
Meta AI app	Meta's companion app that pairs, registers and updates the glasses
UI profile	Versioned configuration of what a pair of glasses shows and allows
Four eyes	Rule that a second, different person approves what the first verified
Agreement	Whether independent OCR and vision readings of a value match
MoSCoW	Must, Should, Could, Won't prioritisation
Demo mode	Zero-configuration deployment with an in-memory synthetic tenant

16 Appendix: traceability to other documents

Each requirement area is specified in more depth in the other documents of the suite. The table shows where to look.

Table 34. Traceability from requirement areas to the documentation suite

Requirement area	ARCHITECTURE	DATA_MODEL	API_REFERENCE	MOBILE_APP_SPEC	GLASSES_INTEGRATION	D
Identity and tenancy (FR-001 to FR-012)	Tenancy, token model	Identity tables, RLS	Auth, org	Sign-in		
Telemetry (FR-020 to FR-029)	Telemetry flow	Plant tables	Plant, telemetry	Sensor screens		D
Alarms (FR-030 to FR-038)	Alarm flow	Alarm tables	Alarms	Alarm screens		Sc
Work (FR-040 to FR-047)	Offline sync	Work tables	Inspections, maintenance	Work tab	HUD screens	
Devices (FR-050 to FR-059)	Pairing flow	Device tables	Devices, UI profiles	Glasses screen	Capability matrix	H
Android (FR-090 to FR-099)	Containers		All client routes	Whole document	DAT flows	Te
Audit scanning (FR-060 to FR-069)	Audit scan flow	Audit document tables	Audit documents	Scan screens	Camera bridge	Sc
Voice (FR-070 to FR-078)	Voice flow		Assistant	Assistant screen	Speech bridge	V
Documents (FR-110 to FR-114)		Documents	Plant documents			
Platform and insights (FR-080 to FR-089, FR-120 to FR-122)	Realtime	Platform tables	Insights, realtime	Home		C
Website and demo (FR-100 to FR-103)	Demo topology		Public, auth demo	Demo sign-in		W
Non-functional (NFR-001 to NFR-040)	All	Retention	Errors, limits	Performance budgets	Limits	A

NOTE

Every document referenced in this table is part of the suite in `docs/product/` and its Word edition in `docs/product/word/`.